

Single Deal Exposure

- Instrument Valuation
- Case Studies: Demo in Spreadsheet

Modeling Credit Risk Exposure

▶ Trade-Level Calculation

- Credit-Conversion Factor (CCF)
- MtM + CCF Add-on
- Grid Monte-Carlo Simulation

▶ Counterparty-Level Calculation

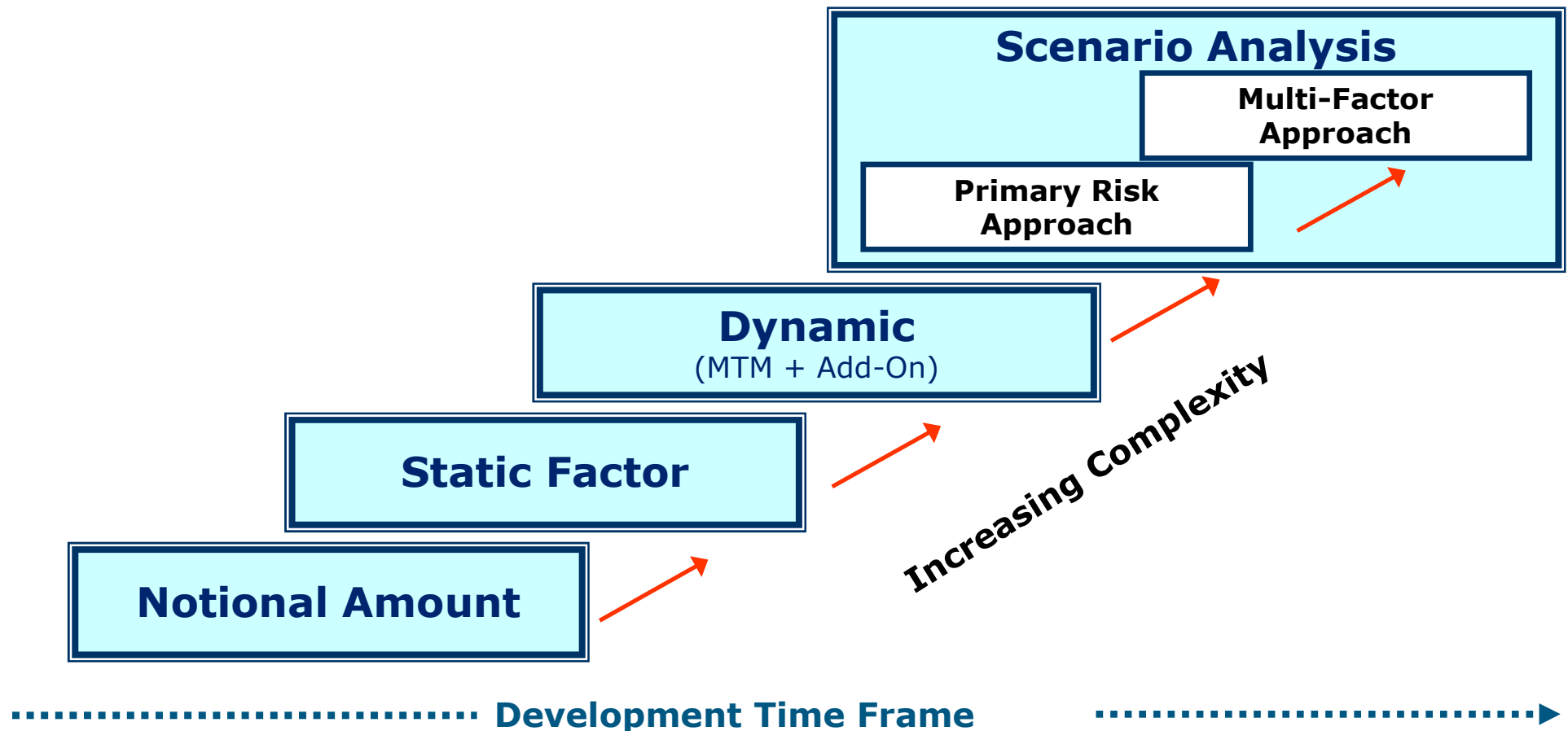
- ISDA Netting Agreement
- Aggregation per Netting set
- Aggregation across Netting sets

▶ Collateralized Exposure Calculation

- ISDA Credit Support Annex (CSA)
- Margin period of risk under CSA
- Modeling the future collateral requirement
- Accounting for threshold and Rating trigger

Evolution of Exposure Measurement Techniques

Most institutions still measure credit exposures based on a mixture between the second and third evolutionary stages.



Exposure Calculation

▶ **MTM Valuation ($T = 0$)**

- Forward: MtM calculated off the FX forward points
- Select: MtM calculated off 3 Points: Near, Far and Mid
- Option: MtM calculated off RR/BtF Implied Vol Surface

▶ **PFE Calculation ($T > 0$)**

- Calculate the exposure profile using the look-up function from a pre-computed Add-on grid
- Calculate the CCY Risk Factor profiles by combining the transaction exposure profiles
- Calculate the Counterparty-level exposure profiles by aggregating the CCY risk factor exposures in Risk Factor simulation.

Mark-to-Market

▶ MTM

- Refers to accounting standards of assigning a value to a position held in a financial instrument based on current fair market price for the instrument or similar instruments.

▶ Forward

- An agreement entered into today to buy or sell a specified asset at a specified future date for an agreed 'forward' price.

▶ Swap

- A forward-type derivative contract whereby one party agrees to exchange cash flows with another party at future dates

▶ Option

- Give the holder the right, but not the obligation, to enter into a financial transaction, at some point in the future, at a predetermined level (strike or exercise price)
 - Call options represent the right to buy. A call option will generally prove profitable for the holder if the price of the underlying asset rises above the agreed strike price
 - Put options are the right to sell. A put option will generally prove profitable if the price of the underlying declines below the strike price

Credit Conversion Factor (CCF) under Basel 2

- *CCF calculates the EAD for derivative transactions.*
- *CCF converts Notional of derivative transaction to its credit exposure*

CCFs proposed by ISDA and Trading Book Review (TBR)

FX	CCF	Equity	CCF	IRP 1yr	CCF	IRP 5yr	CCF
USD/EUR	3.22%	S&P	4.35%	USD	0.19%	USD	0.30%
USD/JPY	2.71%	FTSE	7.36%	EUR	0.14%	EUR	0.19%
USD/GBP	2.79%	DAX	6.62%	JPY	0.05%	JPY	0.13%
EUR/JPY	3.13%	NIKKEI	4.30%	GBP	0.12%	GBP	0.18%
FX (TBR)	2.50%	Equity (TBR)	7.00%	IRP (TBR)	0.20%	IRP (TBR)	0.20%

Credit Conversion Factor (CCF)

► Model Assumption & Calculations

- **Lognormal Distribution**
 - return of the price or rate has a normal distribution
- **The evolution of FX rate follows a Geometric Brownian Motion**

$$\Delta S = \mu S \Delta t + \sigma S z \sqrt{\Delta t}$$

- **CCF Add-on are calculated from**

$$CCF_{UP} = \exp \left[\left(\mu - 0.5 \sigma^2 \right) t + 1.645 \sigma \sqrt{t} \right] - 1$$

$$CCF_{DN} = 1 - \exp \left[\left(\mu - 0.5 \sigma^2 \right) t - 1.645 \sigma \sqrt{t} \right]$$

- **Draft (μ) and Volatility (σ) calculated from Historical Data**

Credit Exposure for FX Trades

▶ Add-on Calculation

- Create CCF Add-on based on Market Volatility

▶ MtM Valuation

- FX Forward: priced off FX forward points
- FX Option: priced using Implied Volatilities.

▶ Transaction-level Calculation

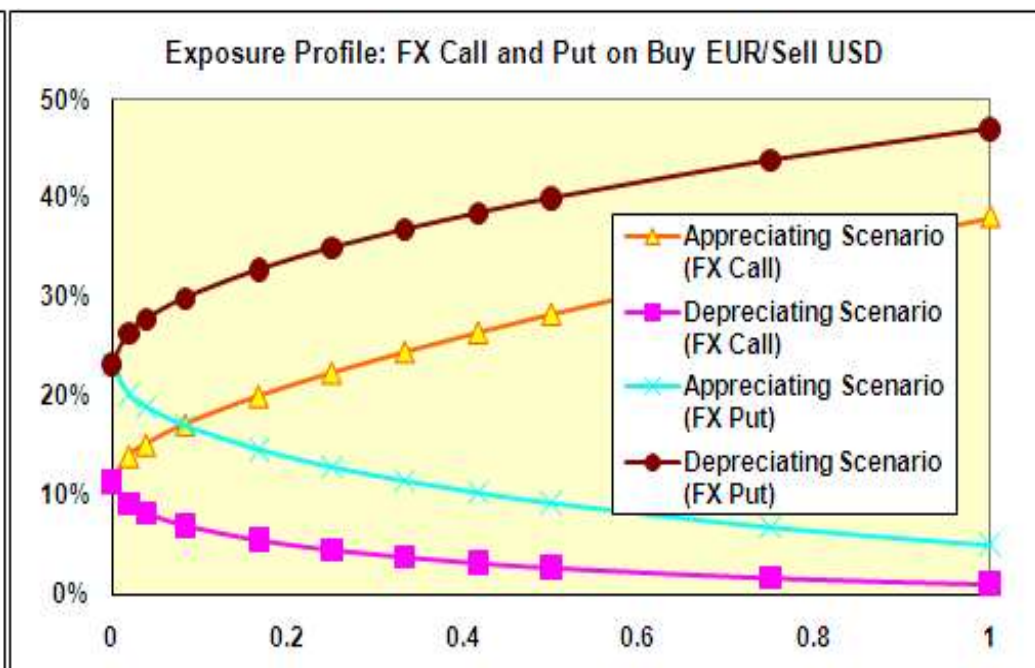
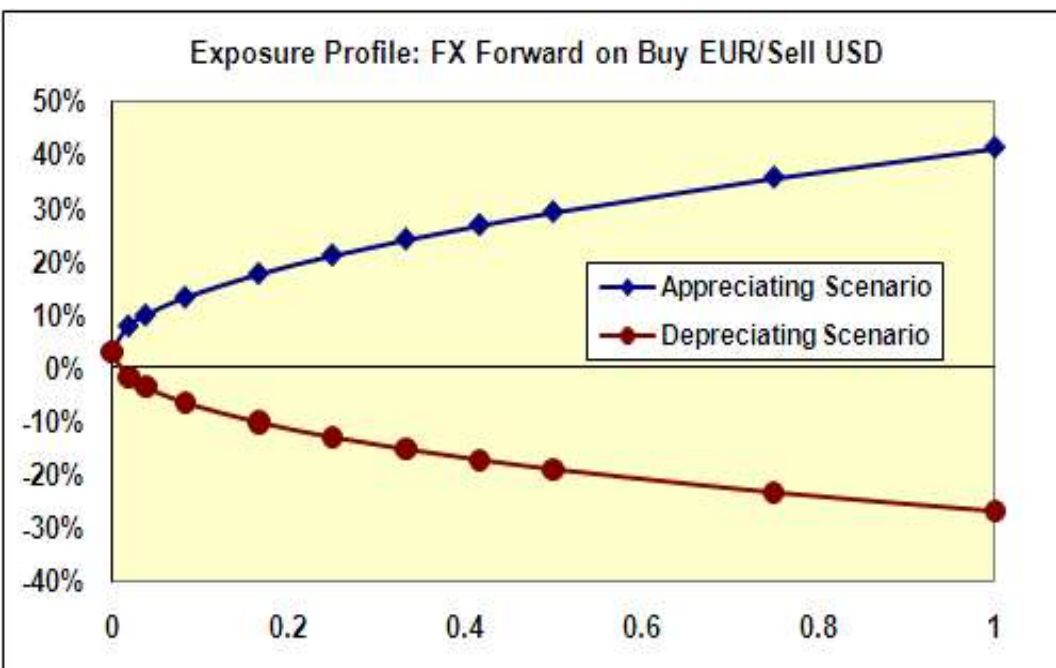
- Apply “MtM + Add-on” method to generate Exposure profiles at Up, Mid and Down scenarios
- Cross-currency transactions are split into 2 transactions

▶ Customer-level Exposure Calculation

- Monte-Carlo simulation to ensure scenario consistency
- Aggregation across CCY RF with Netting or Non-netting

MtM + Add-On Method

- ▶ **Appreciating Scenario** 升值情境 ($T > 0$)
 - $CRU(T) = MtM + Add-On(T, Rec.CCY)$, for Pay.CNY/Rec.CCY trade
 - $CRU(T) = MtM + Add-On(T, Pay.CCY)$, for Rec.CNY/Pay.CCY trade
- ▶ **Depreciating Scenario** 贬值情境 ($T > 0$)
 - $CRD(T) = MtM - Add-On(T, Rec.CCY)$, for Pay.CNY/Rec.CCY trade
 - $CRD(T) = MtM - Add-On(T, Pay.CCY)$, for Rec.CNY/Pay.CCY trade



CCF Add-On PE Grid Table

CCY	Vol	Drift	Fwd	0	1 wk	2 wk	1 mo	2 mo	3 mo	6 mo	9 mo	1 yr
AUD	15%	-1.75%	Pay USD	0.0%	3.4%	4.8%	7.1%	10.1%	12.3%	17.4%	21.2%	24.4%
			Rec USD	0.0%	3.4%	4.8%	7.1%	10.0%	12.2%	17.2%	21.0%	24.1%
CAD	10%	-1.50%	Pay USD	0.0%	2.3%	3.2%	4.7%	6.6%	8.0%	11.2%	13.6%	15.5%
			Rec USD	0.0%	2.3%	3.2%	4.8%	6.8%	8.4%	11.9%	14.6%	16.8%
CHF	11%	-2.25%	Pay USD	0.0%	2.5%	3.5%	5.1%	7.2%	8.7%	12.0%	14.5%	16.5%
			Rec USD	0.0%	2.5%	3.6%	5.3%	7.6%	9.3%	13.3%	16.3%	18.9%
EUR	16%	-1.50%	Pay USD	0.0%	3.7%	5.2%	7.6%	10.8%	13.3%	18.8%	23.0%	26.5%
			Rec USD	0.0%	3.6%	5.1%	7.5%	10.6%	12.9%	18.1%	22.0%	25.2%
GBP	13%	0.75%	Pay USD	0.0%	3.0%	4.3%	6.4%	9.1%	11.3%	16.3%	20.3%	23.7%
			Rec USD	0.0%	2.9%	4.1%	6.0%	8.4%	10.2%	14.1%	17.0%	19.3%
HKD	6%	-3.25%	Pay USD	0.0%	1.3%	1.8%	2.6%	3.5%	4.2%	5.4%	6.2%	6.7%
			Rec USD	0.0%	1.4%	2.0%	3.1%	4.5%	5.6%	8.3%	10.5%	12.5%
JPY	15%	0.75%	Pay USD	0.0%	3.5%	4.9%	7.3%	10.4%	12.8%	18.4%	22.8%	26.6%
			Rec USD	0.0%	3.4%	4.8%	7.0%	9.8%	11.9%	16.5%	19.9%	22.7%
SGD	8%	-1.75%	Pay USD	0.0%	1.8%	2.5%	3.7%	5.2%	6.2%	8.6%	10.3%	11.7%
			Rec USD	0.0%	1.8%	2.6%	3.9%	5.6%	6.9%	9.8%	12.1%	14.1%

Interest rate Swap - Valuation

► Interest Rate Swaps

- For pay-fixed swap, MtM is computed as discounted cashflow on difference btw the floating-rate payments and the fixed-rate payments
- $MtM(t) = (Notl = \$100m) \cdot [\sum_{t_i \geq t} F_i(t) b_i d_i(t) - K \sum_{t_i \geq t} b_i d_i(t)]$
 $= (Notl = \$100m) \cdot D(t) \cdot [S(t) - K]$, for $D(t) = \sum_{t_i \geq t} b_i d_i(t) \sim$ Swap Duration
- For receive-fixed swap, MtM is computed as discounted cashflow on difference btw the fixed-rate payments and the floating-rate payments
- $MtM(t) = (Notl = \$100m) \cdot [K \sum_{t_i \geq t} b_i d_i(t) - \sum_{t_i \geq t} F_i(t) b_i d_i(t)]$
 $= (Notl = \$100m) \cdot D(t) \cdot [K - S(t)]$, for $D(t) = \sum_{t_i \geq t} b_i d_i(t) \sim$ Swap Duration



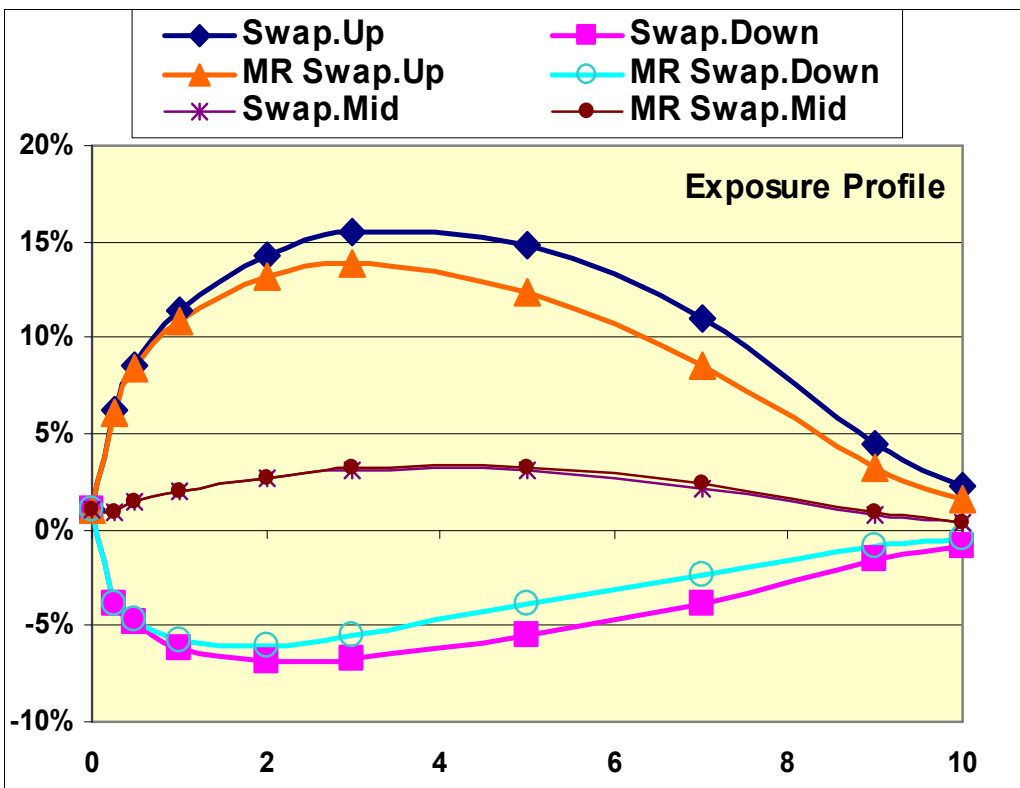
Interest Rate Swap with a Swap Dealer



Exposure Calculation: Swap and Cap

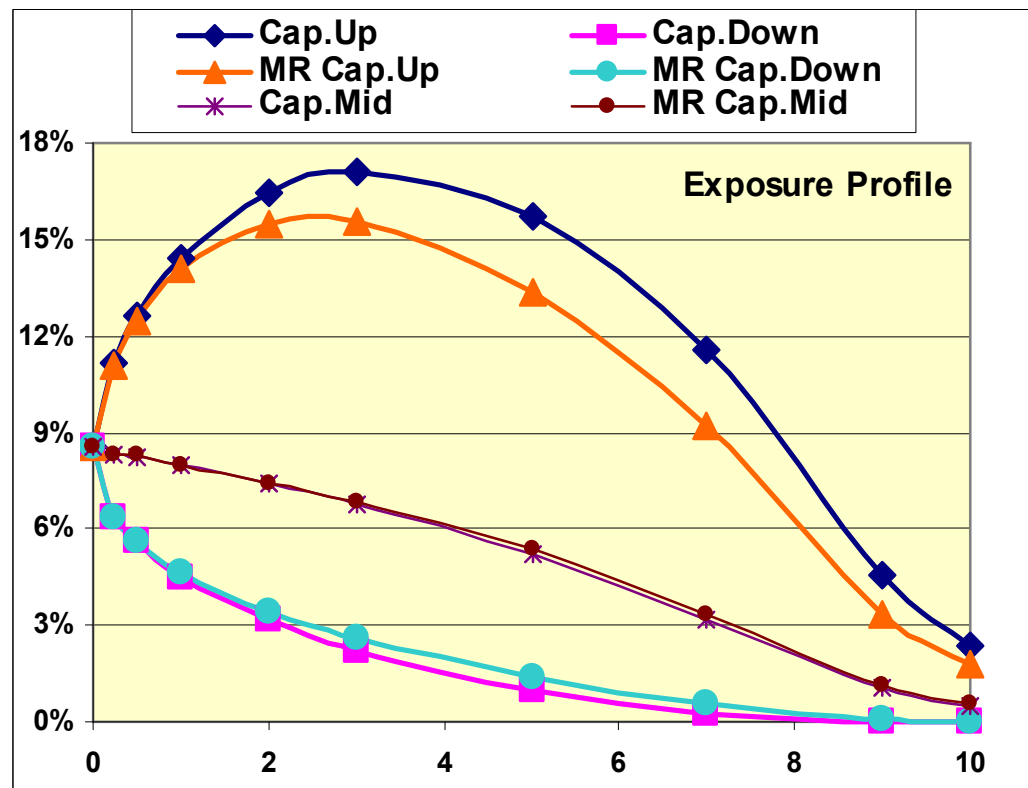
► IR Swaps

- Swap: Up, Mid & Down
- MR Swap: Up, Mid & Down
- Profile: Swap vs MR Swap



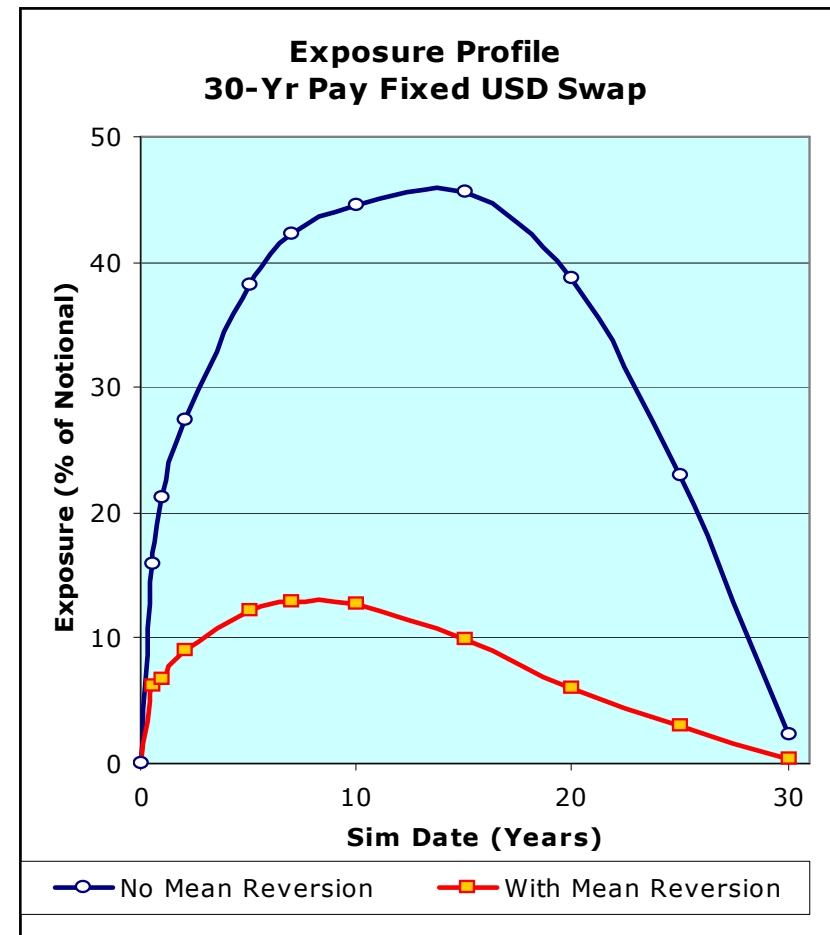
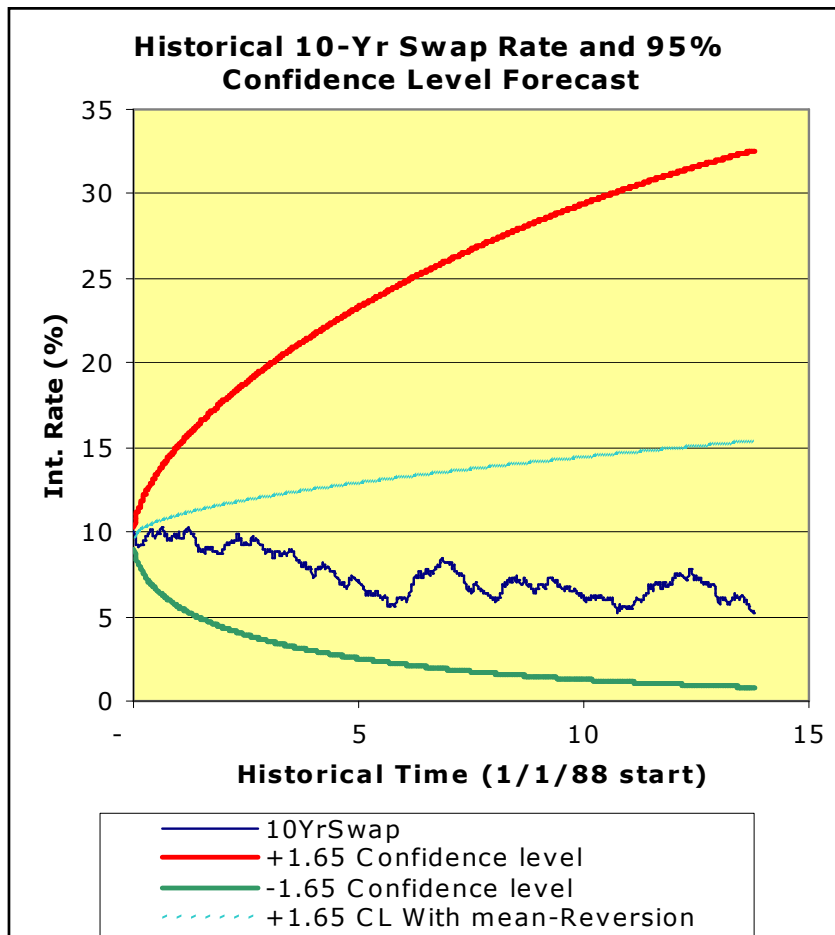
► IR Caps

- Cap: Up, Mid & Down
- MR Cap: Up, Mid & Down
- Profile: Cap vs MR Cap



Effect of Mean Reversion

- **Mean-Reversion** – Tendency of a risk factor to revert back to a long-run mean



How Exposure Evolves Over Time

There are two main factors that determine how PFE varies with time: diffusion and amortization.

► Diffusion

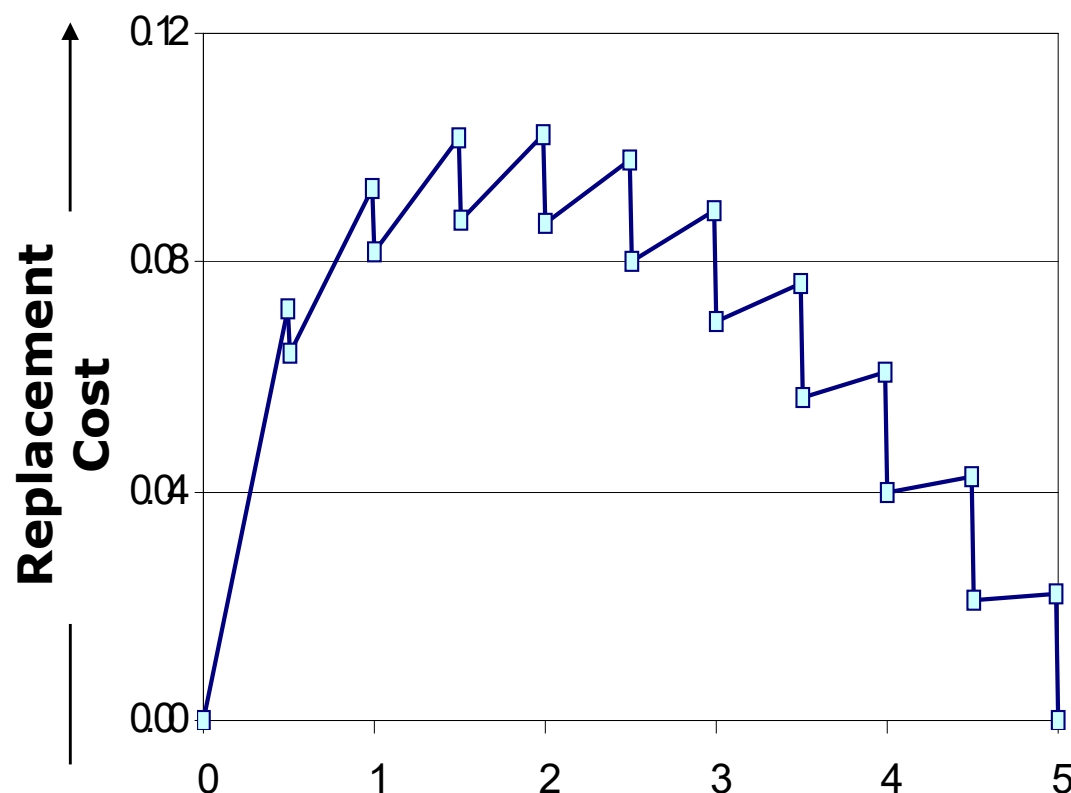
- The “diffusion effect” increase exposure since there is greater potential for rates to move away from current levels

► Amortization

- The “amortization effect” decrease exposure as periodic payments are made reducing the remaining cash-flow exposed to default

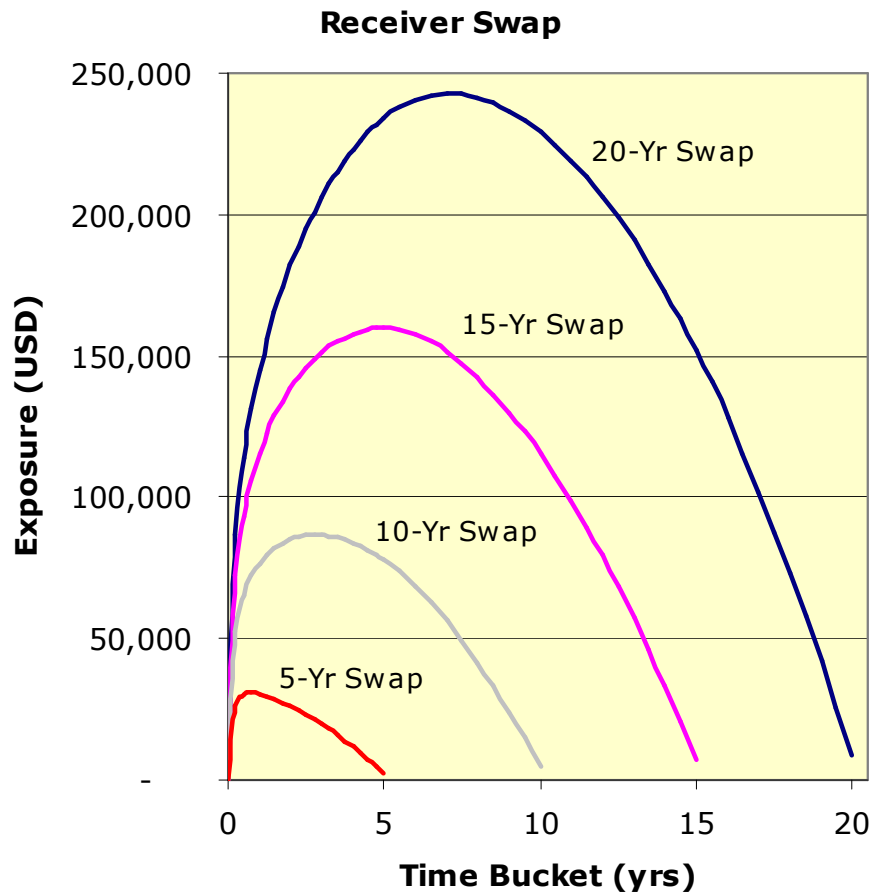
► Multiple Cash Flow Exposure

- As result of Diffusion and Amortization, PFE for interest rate swap usually peaks at about one-third to one-half of the way into the life of the transaction.

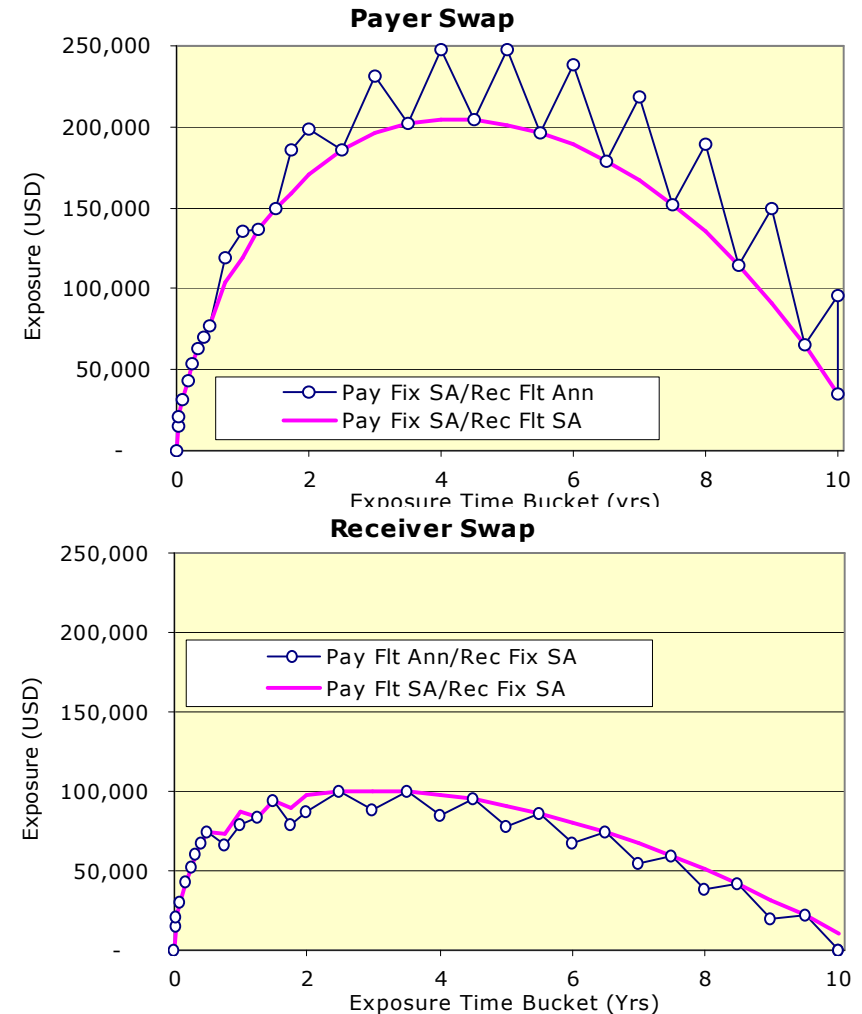


Exposure Profiles of Derivative Products

Effect of Maturity

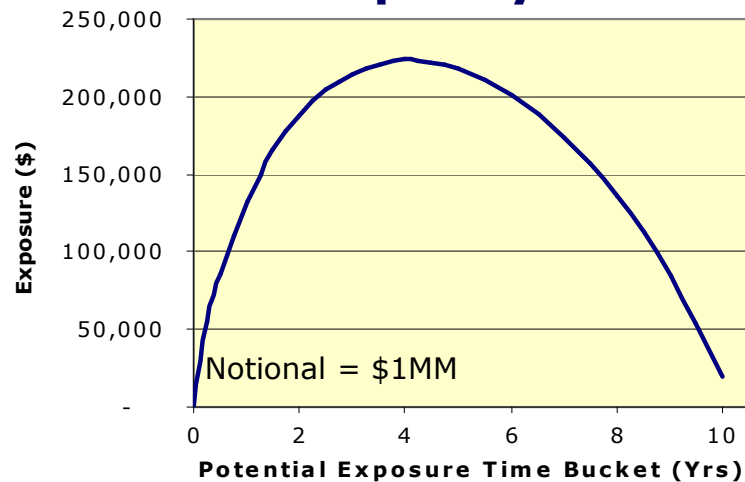


Unequal Payments

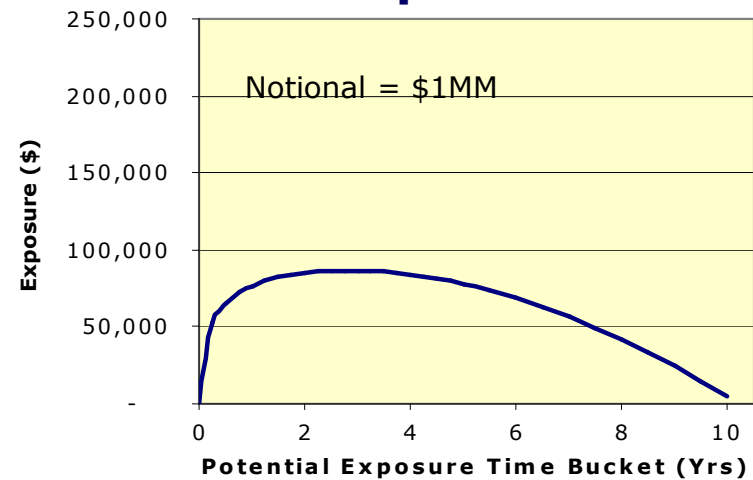


Exposure Profiles of Derivative Products

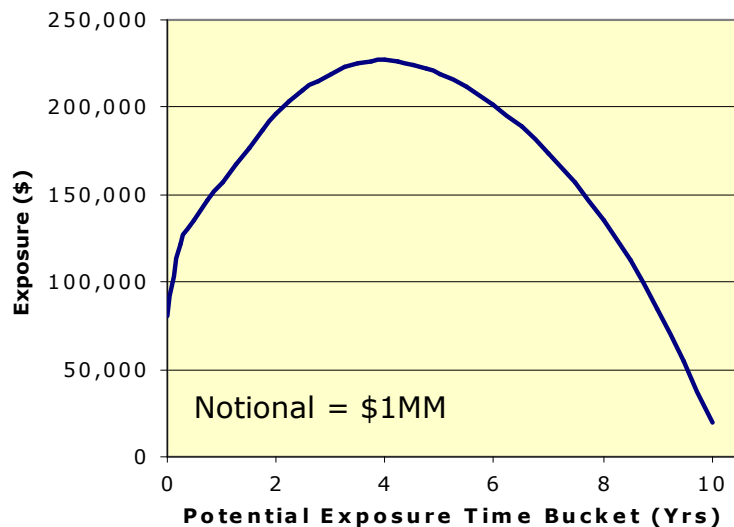
IR Swap: Pay-Fixed



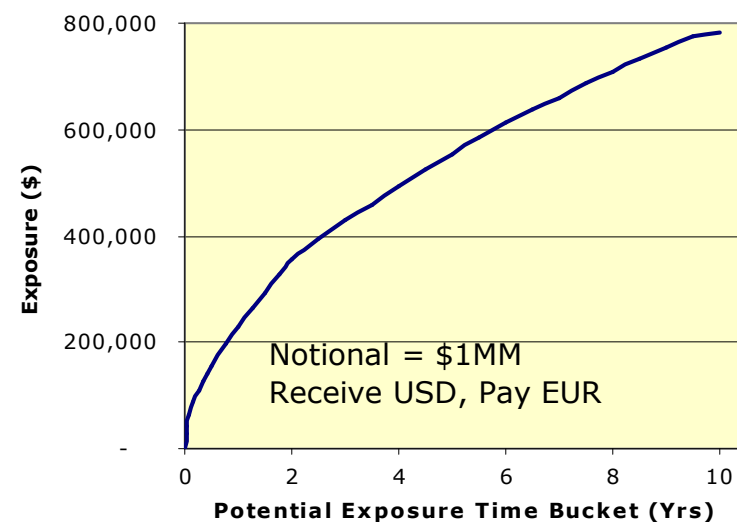
IR Swap: Rec-Fixed



Interest Rate Cap



Currency Swap



Conditional Valuation for Path-Dependent Instruments

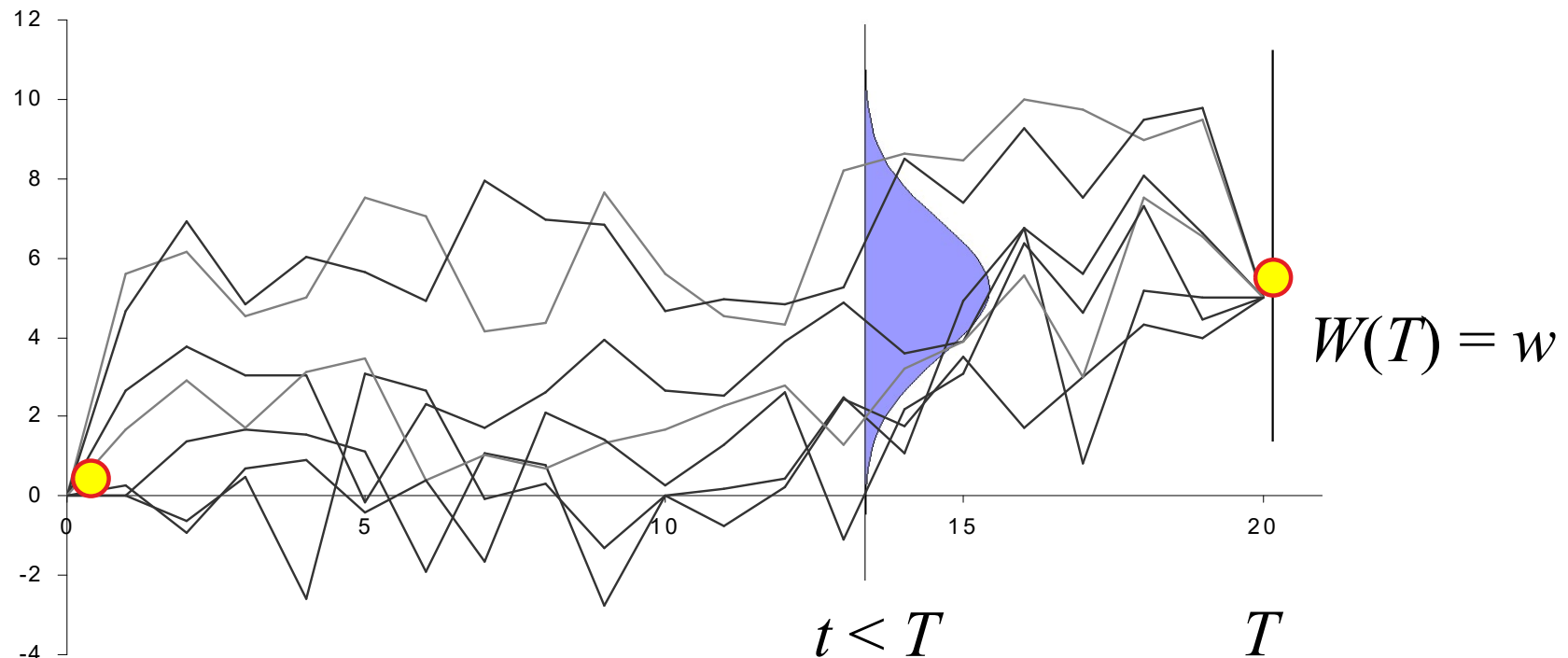
- ▶ The front office models cannot be used for products
- ▶ Scenarios are simulated for a discrete set of dates; derivative value may depend on
 - full continuous paths to the simulation
 - discrete set of dates that may differ from simulation dates (e.g., Bermudan swaptions)
- ▶ Therefore, the *incomplete* information problem arises when using front office models, which rely on a single scenario
- ▶ Conditional Valuation is introduced in Lomibao and Zhu (2005)
 - Path-dependent derivative is valued as expectation of its MTM value **conditional** on all available information from today to the simulation date
 - Mathematically, this is accomplished via **Brownian Bridge**

Exotic Derivatives QUANTITATIVE FINANCE COHORT

Derivative	Description	Payoff Function
Barrier Option	Option that activates or expires if a certain price level is reached.	Payoff = $(S_T - K)^+$ if barrier condition is met.
Asian Option	Option with payoff based on the average price of the underlying asset.	Payoff = $\max(0, S_{\text{avg}} - K)$.
Lookback Option	Option with payoff based on the maximum or minimum price of the asset.	Payoff = $\max(S_T - S_{\min}, 0)$.
Digital Option	Binary option that pays a fixed amount if a condition is met.	Payoff = $\begin{cases} P & \text{if } S_T > K \\ 0 & \text{if } S_T \leq K \end{cases}$.
Cliquet Option	Option with periodic resetting and locking in gains.	Payoff = $\sum \max(S_i - K_i, 0)$.
Rainbow Option	Option based on the best or worst performer of multiple assets.	Payoff = $\max(S_1 - K_1, S_2 - K_2)$.
Chooser Option	Option where the holder can choose between a call or put at a later date.	Payoff = $\max(S_1 - K, K - S_T)$.
Worst-of Option	Option based on the worst performing asset in a basket.	Payoff = $\min(S_1, S_2) - K$.

Concept of Conditional Pricing

- ▶ Mathematically, it is known as “Brownian Bridge”
- ▶ It is about what may have happened in the past given the future state
- ▶ The problem is to determine the distribution of $W(t)$ at any time conditional on knowing the end points $W(0) = w_0$ and $W(T) = w$



Conditional Pricing - Brownian Bridge

- **Conditional on $W(T)=w$, the density of Brownian bridge is given by**

$$f(W(t)=x | W(T)=w) = \frac{1}{\sqrt{2\pi \cdot t(1-\rho^2)}} \exp\left[-\frac{(x/\sqrt{t} - \rho w/\sqrt{T})^2}{2(1-\rho^2)}\right]$$

- **Property of Brownian bridge:**

- $E[W(t)|W(T)] = (t/T) \cdot W(T)$
- $\text{Var}[W(t)|W(T)] = t(1-t/T)$

where $\rho = \sqrt{t/T}$, $0 < t < T$ and $W(t)$ is a Brownian motion.

Conditional Valuation

► Formulation

$$V(t_k, x) = \mathbb{E} \left[f \left(t_k, \{X(t)\}_{0 \leq t < t_k} \right) \middle| X(t_k) = x \right]$$

where $X(t_k) = x$ is a future scenario, and $\{X(t): 0 \leq t < t_k\}$ denotes entire path of risk factor evolution

► Valuation Function

– $f \left(t_k, \{X(t)\}_{0 \leq t < t_k} \right)$ may be path-dependent

Barrier Option: MtM Valuation

- ▶ **Barrier option is an option that pays off at the expiry if the underlying asset has not breached the barrier level during the term.**

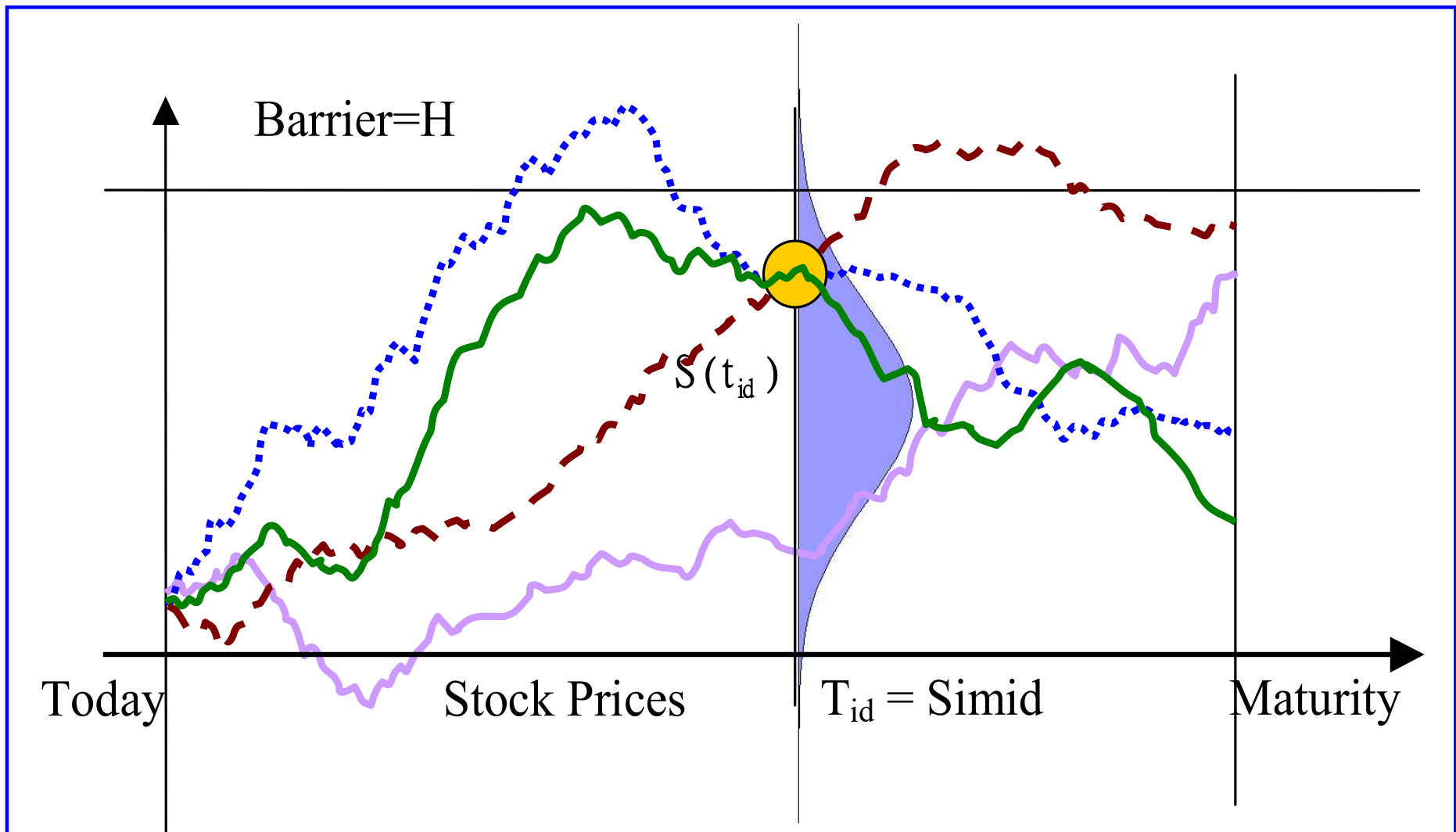
- ▶ **UpOut Call Option**

$$\text{MtM}(0) = E \left[\text{Max} \left(0, S(T_e) - K \right), \left\{ S(t) < H, t < T_e \right\} \right]$$

where H = barrier, K = strike and S(T_e) is the underlying price at T_e = expiry.

Barrier Option: Value-at-Future

- ▶ Which path survived without hitting the barrier?



Barrier Option: Value-at-Future

► **At the future time = T_{id} :**

$$\begin{aligned} VaF(t_{id}, x) &= E \left[MtM(t_{id}, x), h^x(t < t_{id}) \mid S(t_{id}) = x \right] \\ &= MtM(t_{id}) \cdot \text{Prob} \left[S(t) < H, t < t_{id} \mid S(t_{id}) = x \right] \end{aligned}$$

where

$$h^x(t < t_{id}) = \{S(t) < H, t < t_{id}\}$$

► **$MtM(T_{id})$ = the Barrier option formula.**

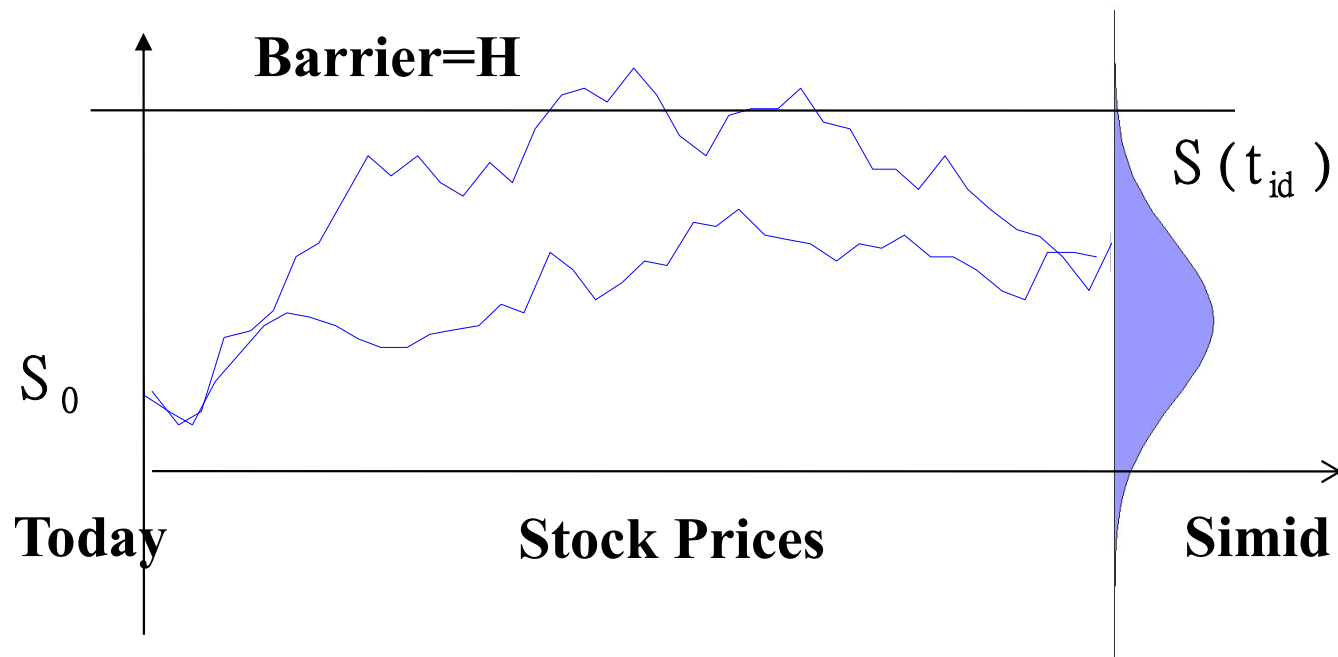
Barrier Crossing Probability

► Brownian bridge crossing the barrier:

$$\text{Prob}[S(t) < H, t < t_{id} \mid S(t_{id}) = x]$$

$$= 1 - \exp\left\{2h(z_{id} - h) / (\sigma^2 t_{id})\right\}, \quad z_{id} \leq h$$

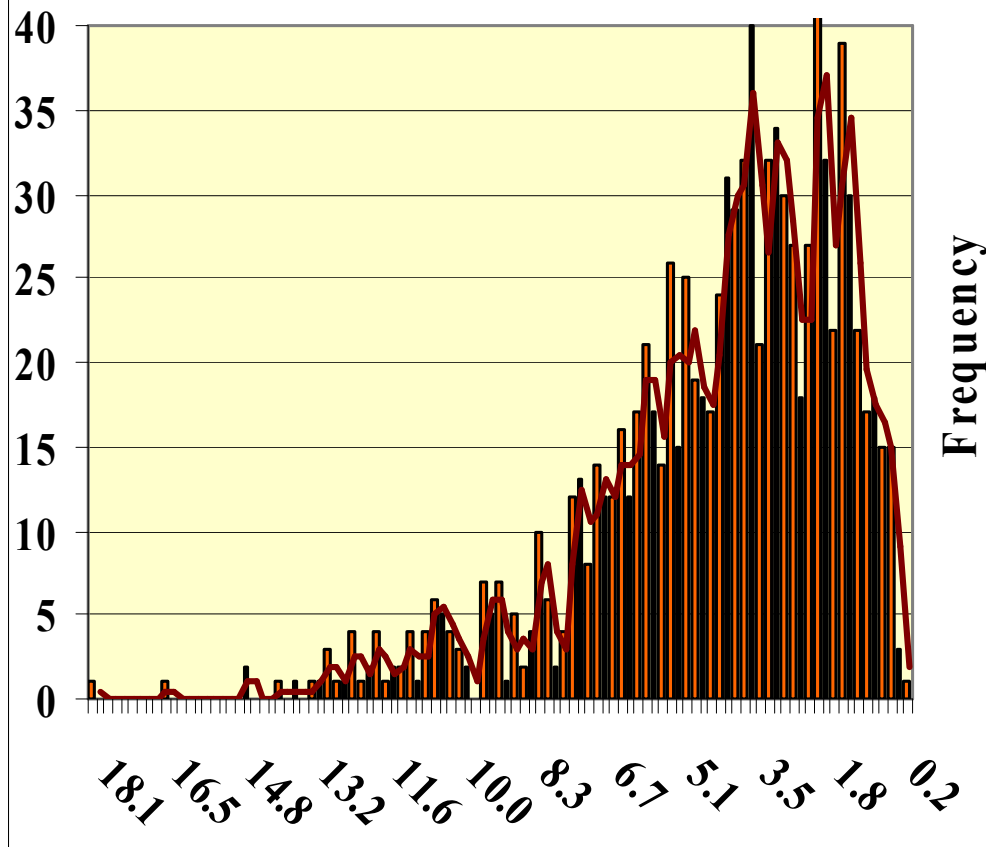
where $z_{id} = \ln[S(t_{id}) / S_0]$ **and** $h = \log(H / S)$



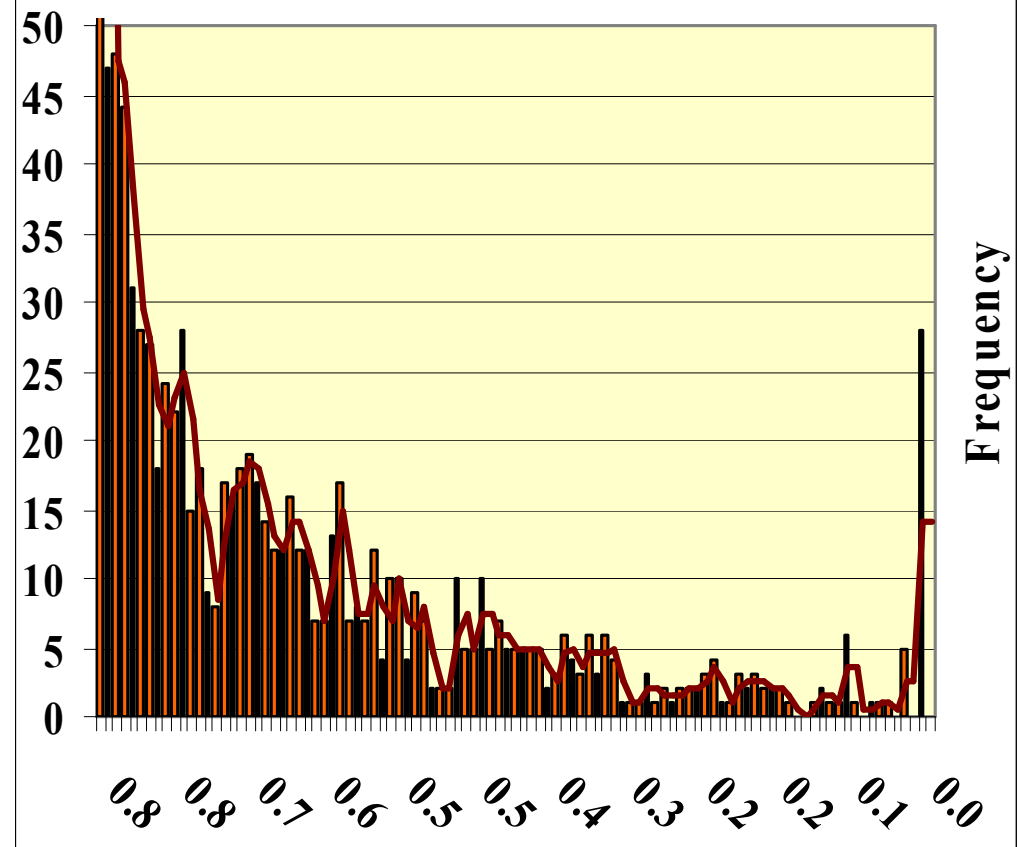
Distribution of Value-at-Future

--- 1000 simulation at $T = 6\text{mo.}$

Vanilla Option (1yr ATM)



Barrier Option (1yr ATM)



Example: Interest Rate Swaption

- ▶ **An option to enter a swap**

- RTP: right to pay the fixed rate
- RTR: right to receive the fixed rate

- ▶ **Payoff at expiry of option**

$$V(T_e) = \text{Max}\{0, \text{Swap}(T_e)\}$$

$$\text{Swap}(T_e) = \sum b_i d_i [F(T_e, T_i) - K]$$

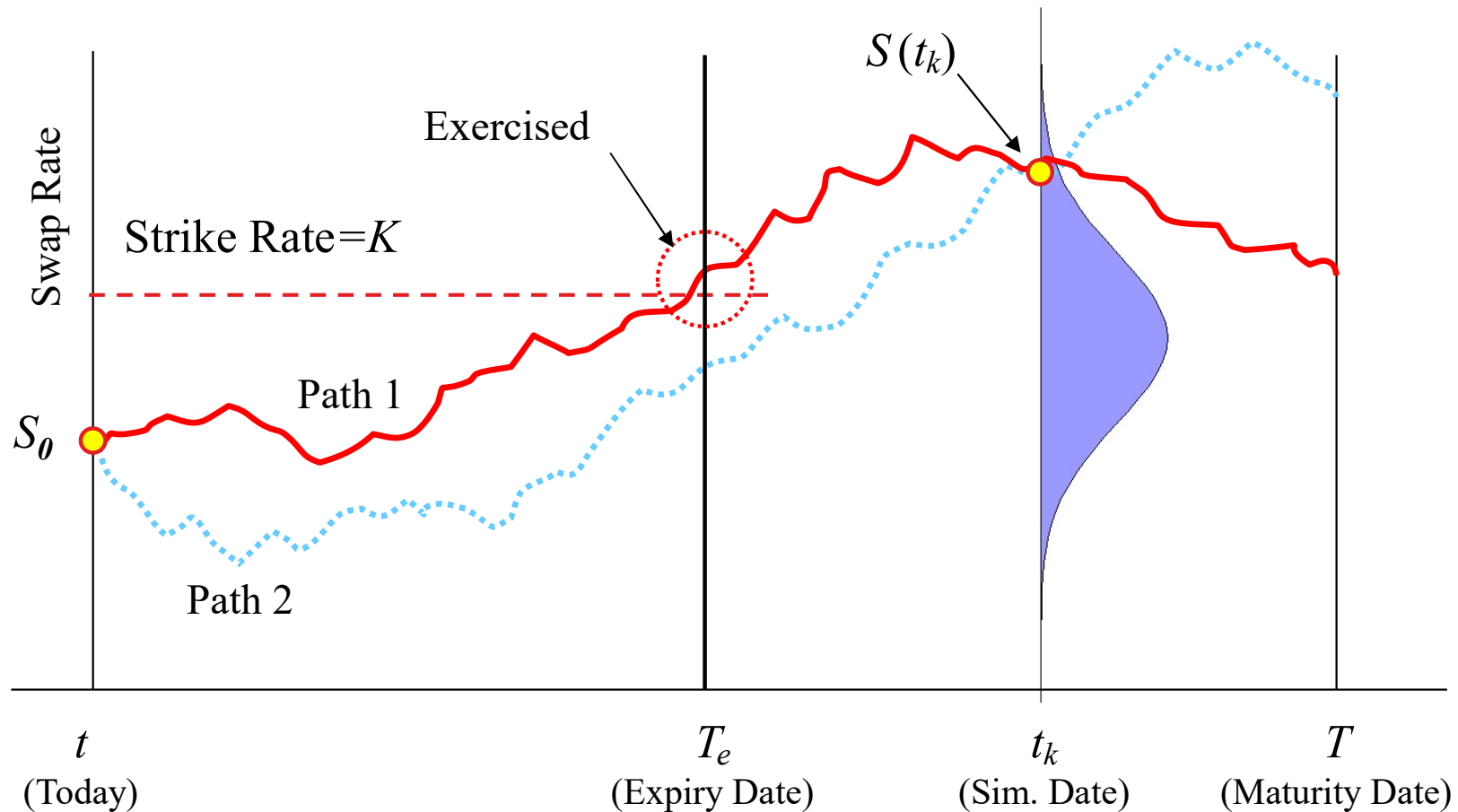
- ▶ **Rewrite in term of swap rate**

$$V(T_e) = (\sum b_i d_i) \times \text{Max}\{0, S(T_e, T_m) - K\}$$

T_e = expiry of option, T_m = maturity of swap

Example: Swap-Settled Swaption

$$\bar{V}_{swaption}(t_k) = \Pr[S(T_E) > K | S(t_k)] \times V_{swap}(S(t_k), t_k, T)$$



Conditional Valuation - Swaption

- **Conditional on swaption exercise at $T_E < t_k$, the future value of a swap-settled swaption is**

$$\bar{V}_{swaption}(t_k) = \Pr[S(T_E) > K | S(t_k)] \times V_{swap}(S(t_k), t_k, T)$$

$\Pr[S(T_E) > K | S(t_k)] =$ **Probability that swaption is exercised into the underlying swap at option expiry T_E**

$V_{swap}(S(t_k), t_k, T) =$ **Value of remaining portion of underlying swap from time t_k to its maturity T given that the simulated swap rate is $S(t_k)$**

Probability of Exercise

- Using Brownian-Bridge properties, the probability of exercise can be easily solved:

Swap rate at expiry time T_E

$$S(t_k) = S(T_E) \cdot \exp \left[\left(\mu - \frac{1}{2} \sigma^2 \right) (t_k - T_E) + z \cdot \sigma \sqrt{t_k - T_E} \right]$$

Probability of exercise at expiry time T_E

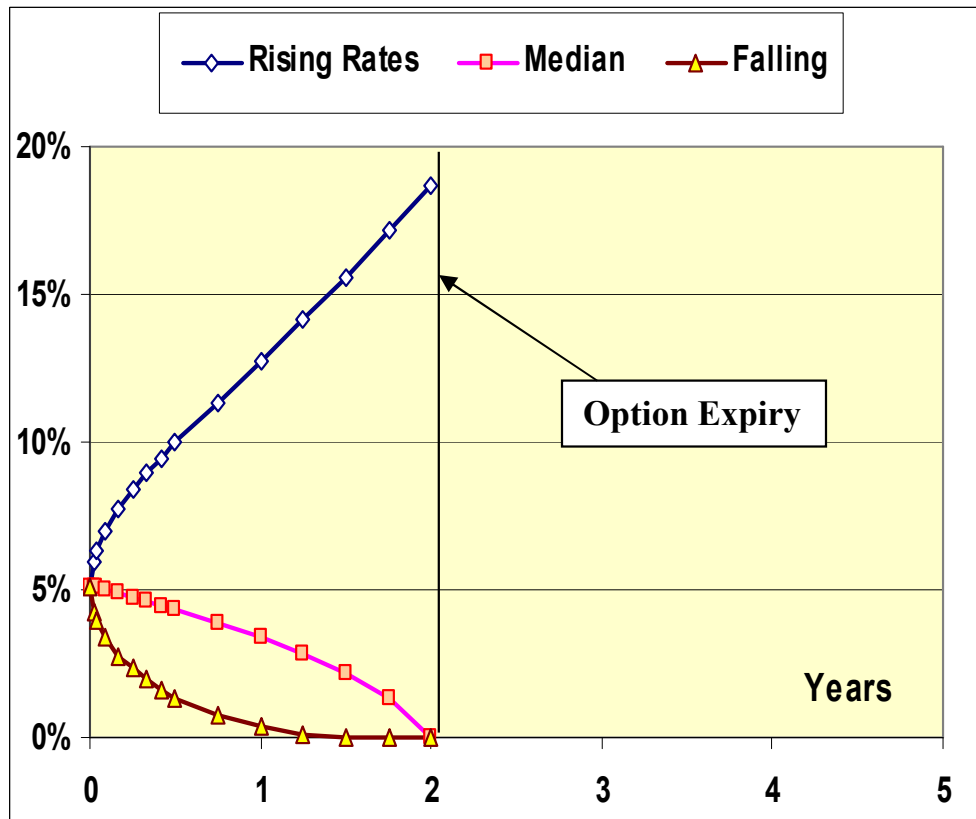
$$\text{Prob} \{ S(T_E) > K \mid S(t_k) \} = 1 - N \left[z^* (T_E, t_k) \right]$$

$$z^* (T_E, t_k) = \frac{\left\{ \ln [K / S_0] - (T_E / t_k) \ln [S(t_k) / S_0] \right\}}{\sigma \sqrt{T_E (1 - T_E / t_k)}}$$

Conditional Valuation - Swaption

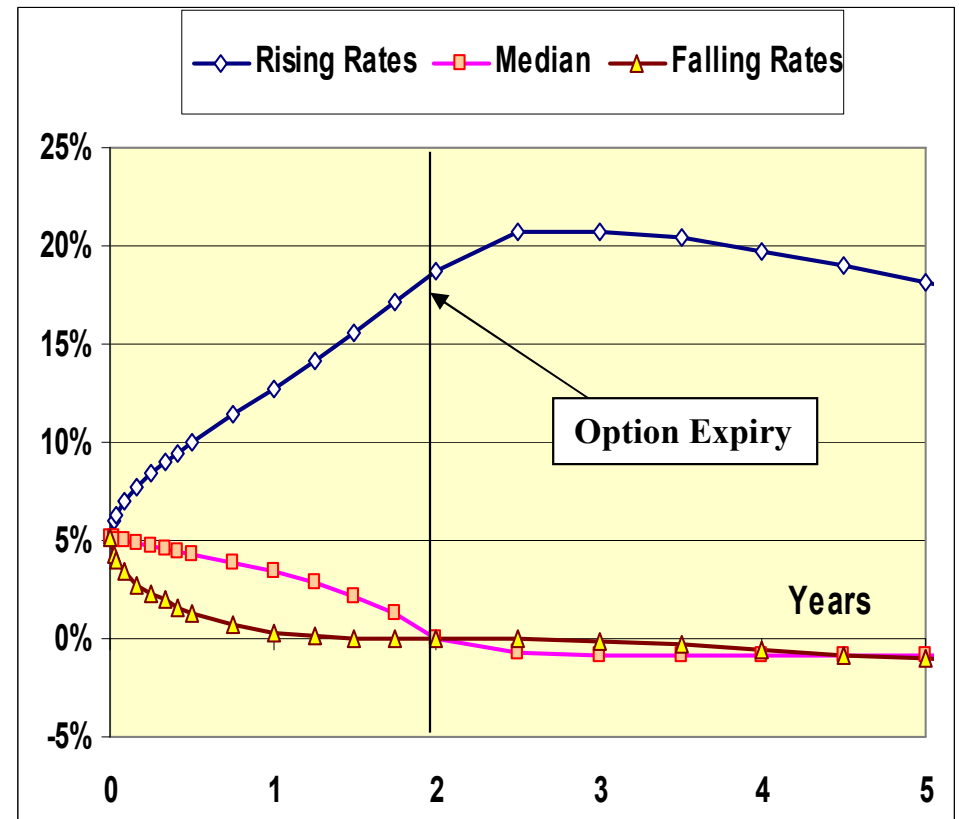
► Cash-settled

- 2yr Option into 10yr Swap



► Swap-settled

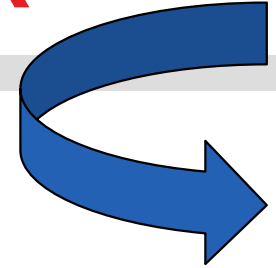
- 2yr Option into 10yr Swap



Exposure to Counterparty

- Aggregating Multiple Deals
- Case Studies: Demo in Spreadsheet

PFE Model Simulation Framework



► Scenario Generation

- Scenario model to generate future market scenarios
- The future scenarios of relevant risk factors need to be correlated

► Instrument Valuation

- Valuation model to evaluate transaction for each scenario
- Using simulated market risk factors as input to pricing models

► Portfolio Aggregation

- Use appropriate netting rules defined in ISDA
- Apply collateral, and other credit enhancements

► Calculate the exposure measures

- Confidence level exposure (e.g., 95%)
- Expected Exposure and Effective Exposure

Aggregation: Portfolio Effects

► Monte-Carlo simulation captures portfolio effects

$$PFE_{Portfolio} \leq \sum_{Deals} PFE_{Deal}$$

- **Diversification benefits from scenario consistency**
 - Trades not likely to experience simultaneous worst cases
- **Netting**
 - Efficiency through offsets of positive and negative PV's
- **Collateral**
 - Collateral calls usually made at the portfolio level

Approaches in Aggregating Exposures

Exposures can be aggregated using simple or complex methodologies depending on the need for precision or speed.

▶ Sum of Peak Exposures

- Peak exposures are summed up regardless of tenor

▶ Sum of Maximum Exposures Across Time Buckets

- Maximum exposures are added up at each time bucket forming an exposure profile for a counterparty

▶ Monte-Carlo (Scenario Consistent)

- Grid Method. Random draws are used to interpolate/extrapolate market values from a grid of instrument valuations.
 - Example of a grid could be the 3 scenarios (–1.65, 0, +1.65 S.D. movements). Potential replacement values are added up for each simulation (taking netting agreement into consideration) - forming a distribution of values at each time bucket.
- Full Valuation. All instruments are revalued for each random draw.

Approaches in Aggregating Exposures

► Sum of Peak Exposures

- Peak exposures are summed up regardless of tenor

Trade #	Transaction Type		Notional Amount (\$MM)	Tenor (Years)	Peak Tenor (Years)	Peak Exposure (\$MM)
1	JPY Int. Rate Swap	BA Pays Fixed	200	3	2	2.2
2	USD Int. Rate Cap	BA Buys Cap	50	4	2	3.6
3	GBP/USD Currency Swap	BA Receives GBP	10	5	5	2.9
Total Exposure					5	8.7

- Although Trade 1 matures after 3 years and Trade 2 matures after 4 years, the Total Exposure still includes them.
- This type of aggregation is very conservative.

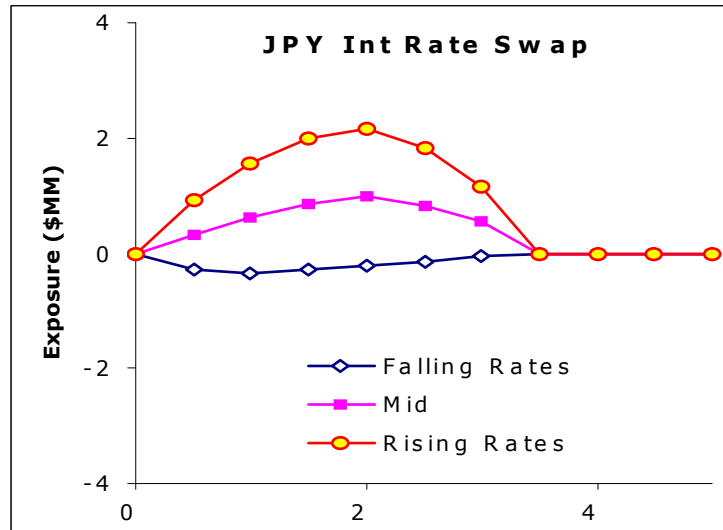
Approaches in Aggregating Exposures

► Sum of Exposures Across Time Buckets

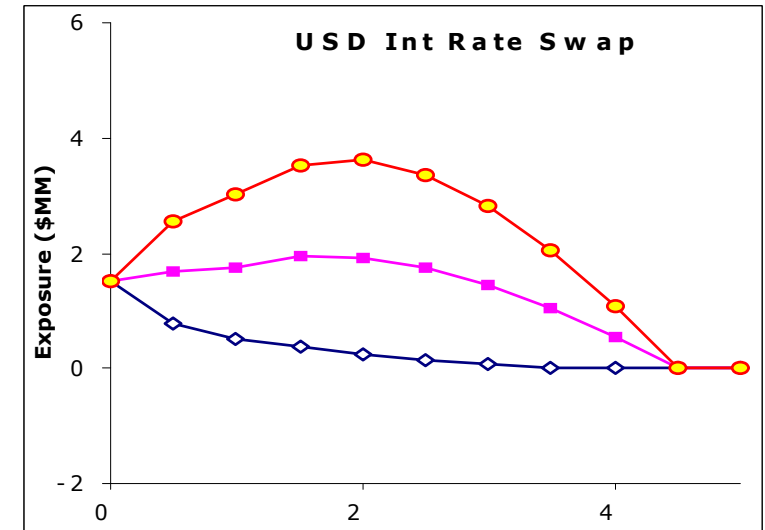
	Exposure Profile (In \$MM)									
Sim Date (Years)	JPY Int Rate Swaps			USD Interest Rate Cap			GBP/USD Currency Swap			Total Exposure
	Falling Rates	Mid	Rising Rates	Falling Rates	Mid	Rising Rates	Falling Rates	Mid	Rising Rates	
0.0	0.0	0.0	0.0	1.5	1.5	1.5	0.0	0.0	0.0	1.5
0.5	-0.3	0.3	0.9	0.8	1.7	2.6	-1.1	0.0	1.0	4.5
1.0	-0.4	0.6	1.6	0.5	1.8	3.0	-1.6	0.0	1.4	6.0
1.5	-0.3	0.9	2.0	0.4	1.9	3.5	-2.0	0.0	1.7	7.2
2.0	-0.2	1.0	2.2	0.2	1.9	3.6	-2.3	0.0	1.9	7.7
2.5	-0.1	0.8	1.8	0.1	1.7	3.3	-2.6	0.1	2.1	7.3
3.0	-0.1	0.6	1.2	0.1	1.4	2.8	-2.9	0.1	2.3	6.3
3.5	-	-	-	0.0	1.0	2.1	-3.1	0.1	2.5	4.6
4.0	-	-	-	-	0.5	1.1	-3.4	0.1	2.6	3.7
4.5	-	-	-	-	-	-	-3.6	0.1	2.8	2.8
5.0	-	-	-	-	-	-	-3.8	0.1	2.9	2.9
Peak Exposure	2.2			3.6			2.9			7.7
Peak Tenor (Yrs)	2.0			2.0			5.0			2.0

Peak

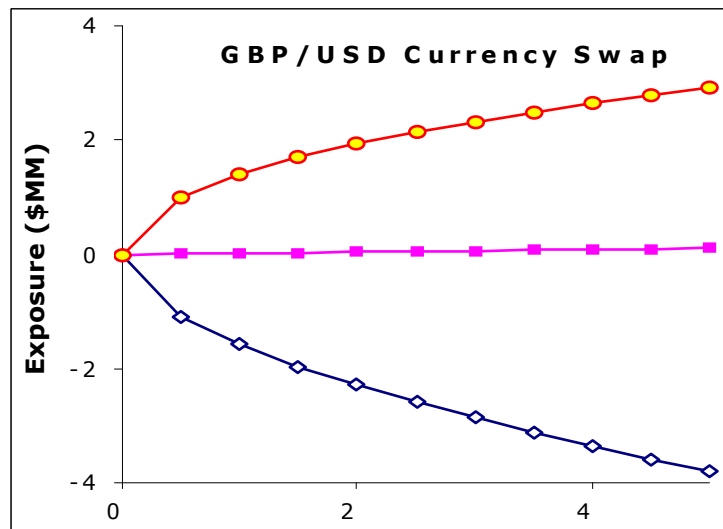
Approaches in Aggregating Exposures



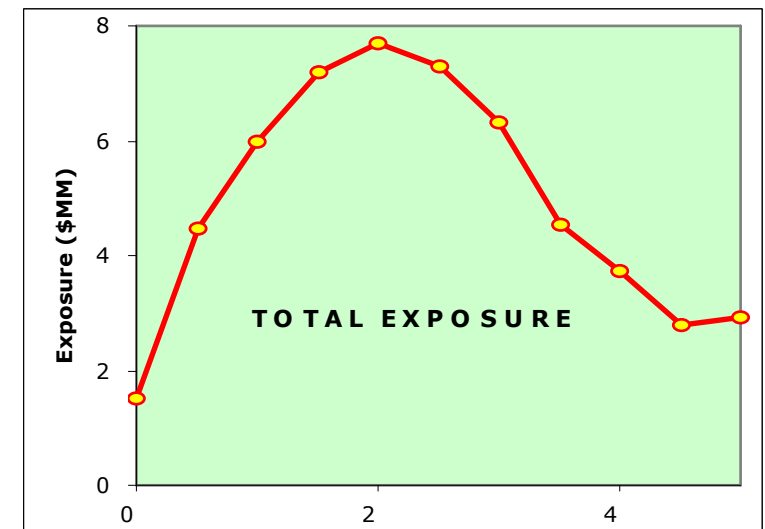
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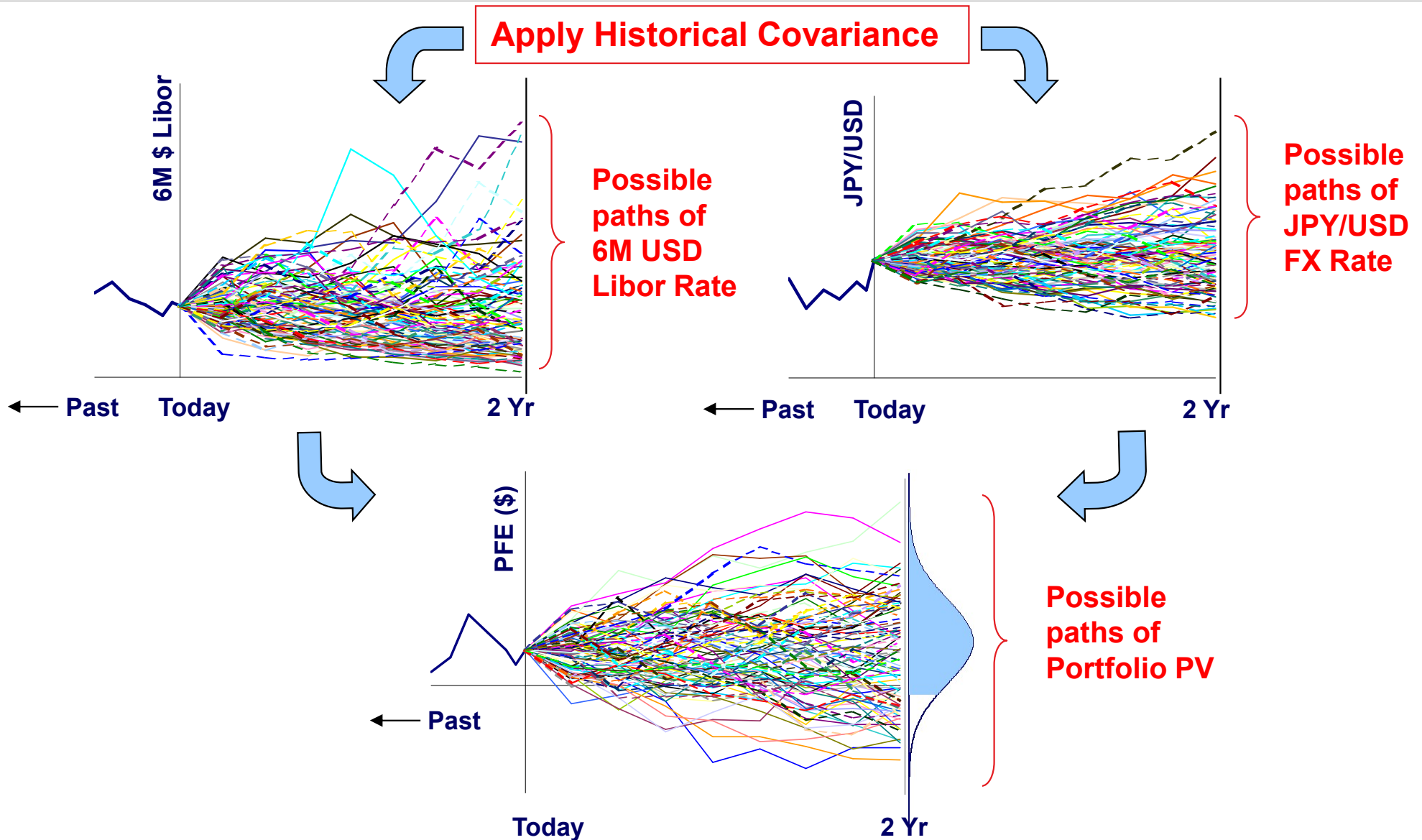
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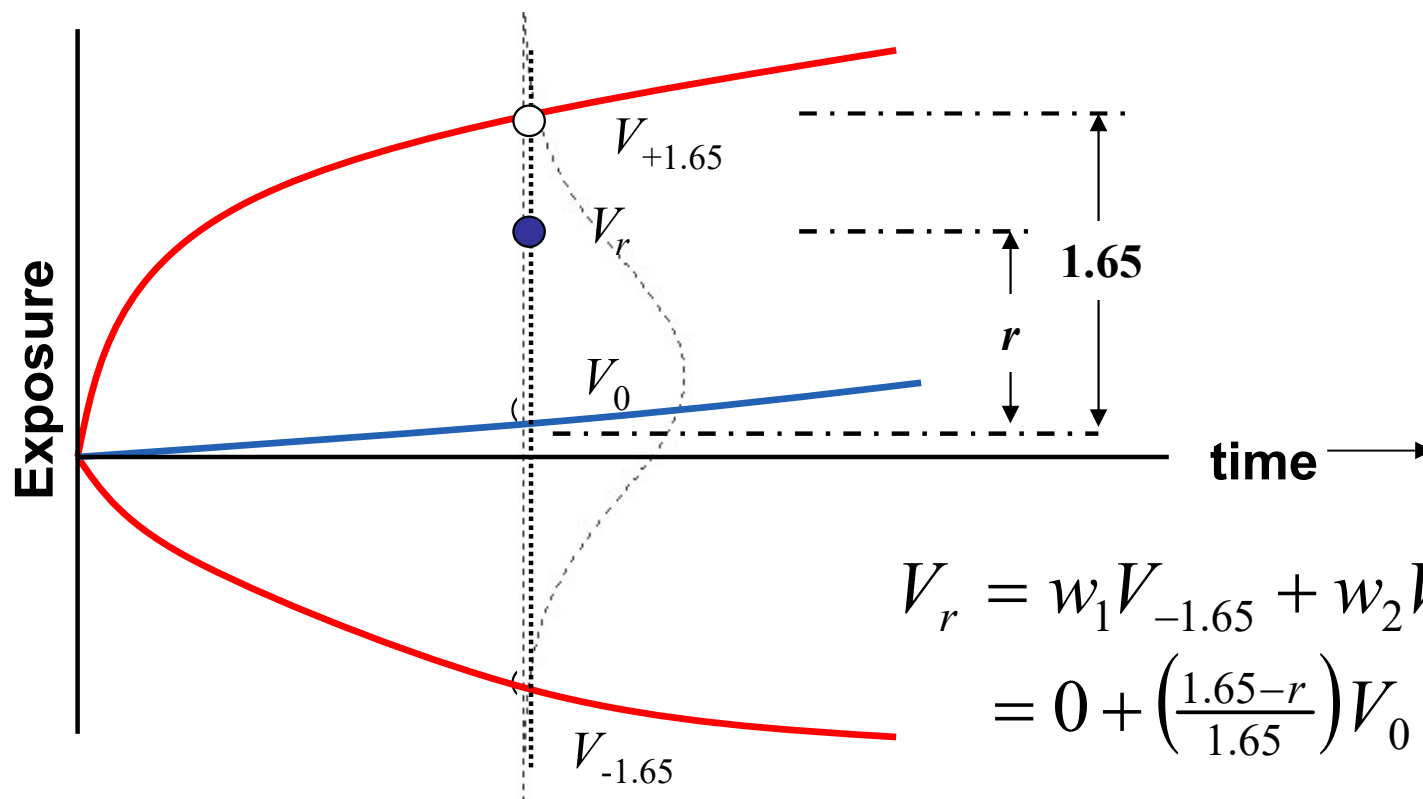


Simulation of Portfolio Aggregation



Grid Monte-Carlo

- ▶ Each MC draw represents number of standard deviation
 - Exposure is interpolated/extrapolated from the 3 scenarios: $V_{+1.65}$, V_0 , $V_{-1.65}$ represent the exposure at the +1.65, 0 and -1.65 standard deviations.

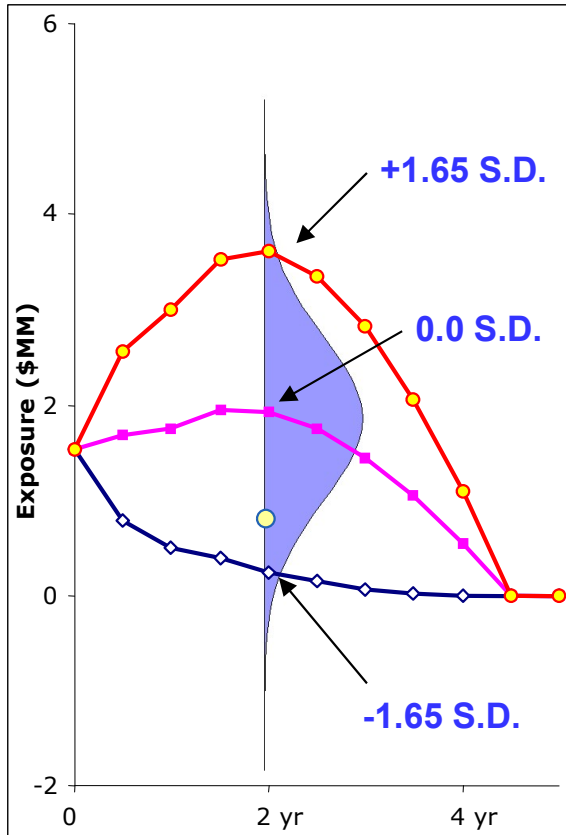


$$\begin{aligned}
 V_r &= w_1 V_{-1.65} + w_2 V_0 + w_3 V_{+1.65} \\
 &= 0 + \left(\frac{1.65-r}{1.65} \right) V_0 + \left(\frac{r}{1.65} \right) V_{+1.65}
 \end{aligned}$$

- Actual Revaluation (Grid Point)
- Interpolated Value (using MC draw)

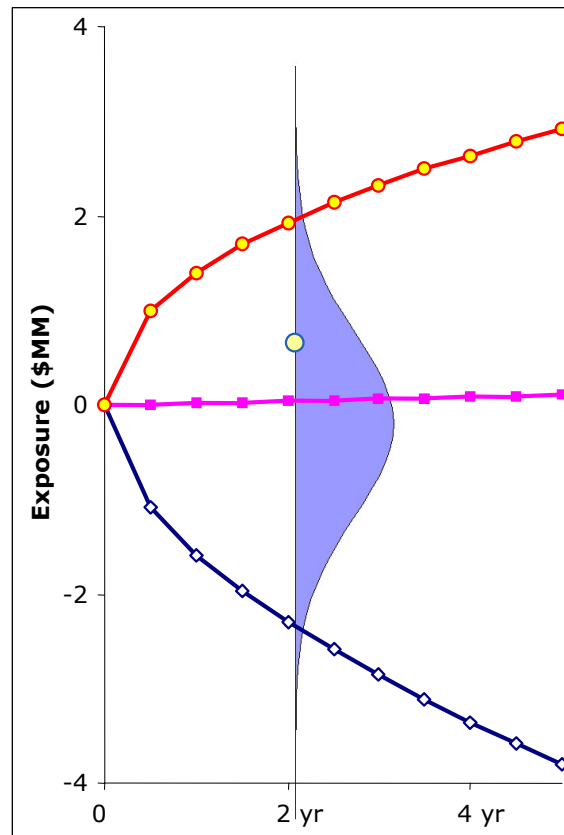
Aggregation Using Grid Monte-Carlo

USD Interest Rate Swap



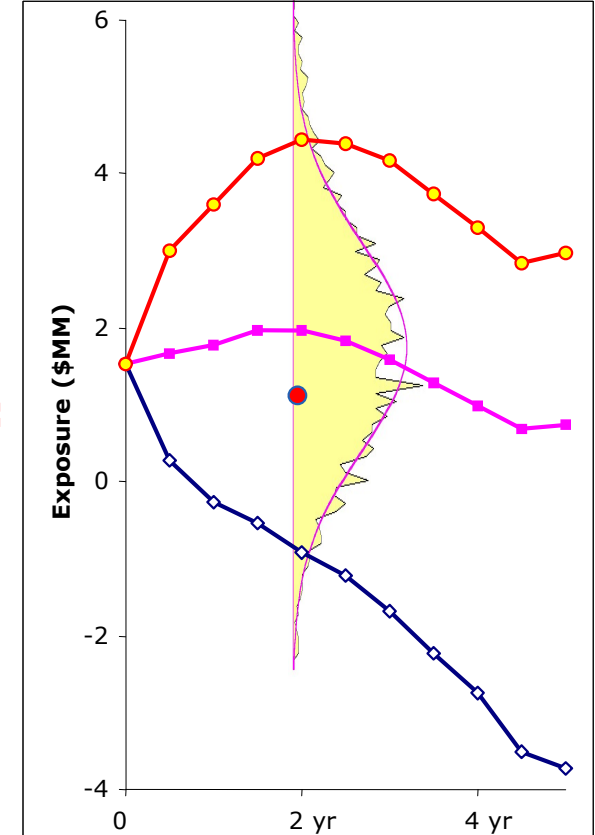
MC Draw: -1.32
Interp Value: \$0.55

GBP/USD Currency Swap



MC Draw: +0.79
Interp Value: \$0.95

TOTAL EXPOSURE



MC Draw: (-1.32, +0.79)
Net Value: \$1.50

Aggregating Across Risk Factors

- ▶ Apply Netting rules for each draw (2000 in total)

$$\begin{array}{c} \text{JPYINT} \end{array} \begin{array}{c} \text{USDINT} \end{array} \begin{array}{c} \text{Risk Factor } n \end{array}$$

$$\begin{bmatrix} V_1^1 \\ V_2^1 \\ \vdots \\ V_{2000}^1 \end{bmatrix} \begin{bmatrix} V_1^2 \\ V_2^2 \\ \vdots \\ V_{2000}^2 \end{bmatrix} \begin{matrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{matrix} \begin{bmatrix} V_1^n \\ V_2^n \\ \vdots \\ V_{2000}^n \end{bmatrix} \xrightarrow{\text{Apply Netting Rules}} \begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ V_{2000} \end{bmatrix} = \overline{\mathbf{V}}$$

- ▶ Sort netted values in ascending order
- ▶ Take the 95th percentile (1,900th item from sorted set)
- ▶ Potential Exposure = max(0, 95th percentile)

Aggregating Across Risk Factors

► Example

- Consider one simulation date, say the 6-month period
- Assume two risk factors (JPYINT, USDINT)
- Two netting agreements
- First MC draw – 1.70 (JPYINT) and +0.80 (USDINT)

Netting Agreement No.	Agreement 1						Agreement 2		
Riskfactor	JPYINT			USDINT			JPYINT		
Deal Number	Deal 101			Deal 102			Deal 103		
Scenario	Falling	Mid	Rising	Falling	Mid	Rising	Falling	Mid	Rising
SimId 15	100	120	160	120	70	-110	-60	-40	20

Interpolation

$$\text{Deal 101: } V_{101} = 120 + (100 - 120) \times (1.70/1.65) = 99.4$$

$$\text{Deal 102: } V_{102} = 70 + (-110 - 70) \times (0.80/1.65) = -17.3$$

$$\text{Deal 103: } V_{103} = -40 + (-60 + 40) \times (1.70/1.65) = -60.6$$

Netting

Since Deals 101 and 102 have the same agreement number, we can net them, i.e., $99.4 - 17.3 = 82.1$

The total exposure derived from this MC draw

$$V = \max(82.1, 0) + \max(-60.6, 0) = 82.1$$

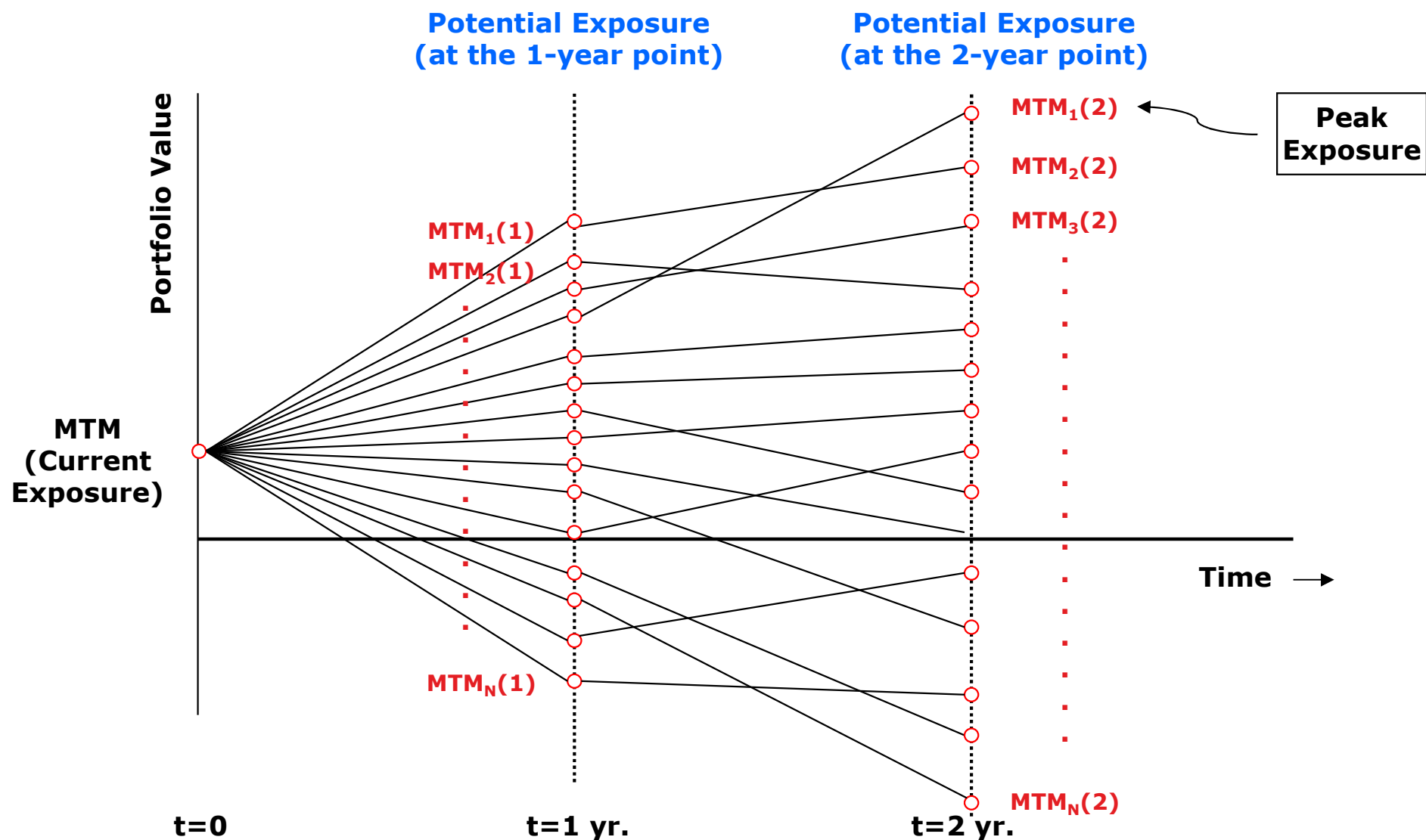
Aggregating Across Risk Factors

► Example: Using 20 Draws

- Assume JPYINT and USDINT are nettable
- GBP/USD trades are in separate netting agreement

N-Draw	JPYINT	USDINT	Apply Netting Rules	Sub-Total	GBP/USD	$\max(\text{Sub-total}, 0) + \max(\text{GBP/USD}, 0)$	TOTAL	Sort in ascending order	TOTAL	} 95th Percentile
1	0.66	2.55		3.20	-0.87		3.20		1.04	
2	0.43	2.12		2.55	-0.51		2.55		1.11	
3	0.39	2.21		2.61	0.34		2.95		1.29	
4	-0.03	1.37		1.34	-0.84		1.34		1.33	
5	0.18	1.70		1.88	-0.16		1.88		1.34	
6	0.85	2.43		3.28	0.39		3.67		1.37	
7	0.40	0.96		1.36	0.49		1.85		1.50	
8	-0.03	1.40		1.37	-0.77		1.37		1.85	
9	1.32	1.53		2.85	-0.20		2.85		1.88	
10	-0.14	1.42		1.29	-0.41		1.29		1.93	
11	-0.32	1.43		1.11	-0.72		1.11		2.25	
12	0.91	1.72		2.63	0.56		3.19		2.28	
13	0.46	0.86		1.33	-0.53		1.33		2.36	
14	0.72	1.52		2.25	-0.91		2.25		2.55	
15	-0.04	2.31		2.28	-0.04		2.28		2.85	
16	0.69	2.51		3.20	-0.67		3.20		2.95	
17	-0.31	1.81		1.50	-0.17		1.50		3.19	
18	0.34	1.73		2.07	0.29		2.36		3.20	
19	0.16	1.77		1.93	-0.82		1.93		3.20	
20	-0.29	1.23		0.94	0.10		1.04		3.67	

Credit Exposure Distribution



Credit Exposure Risk Reports

► Generate Risk Reports

- Key Risk Drivers
- Counterparty Exposure Profile
- Aggregated Exposure Reports
- Exposure Reports by Industry, Region and Risk-Rating.

■ Exposure Profile

时间点	95%置信度	50%置信度	5%置信度
0	6,237	6,237	6,237
1 wk	11,052	5,989	2,285
2 wk	13,347	5,911	1,178
1 mo	16,657	5,731	639
2 mo	20,736	5,308	519
3 mo	23,695	4,863	465
4 mo	26,304	4,773	440
5 mo	28,244	4,542	410
6 mo	29,558	4,151	263
9 mo	23,247	2,403	0
1 yr	9,619	896	0

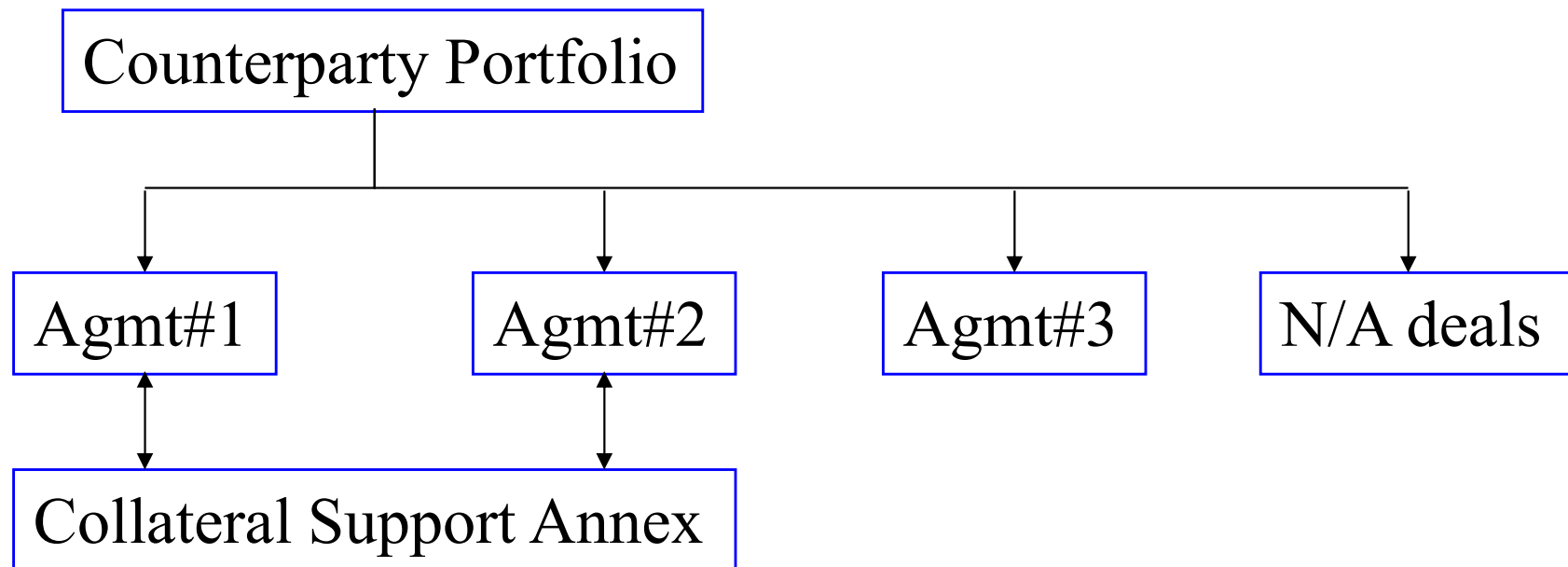
■ CE, 2wk and Peak Exposure %

客户评级 (Cpty Rating)	当前敞口 (Current Exposure)	当前敞口占比 (Current Exposure %)	两周潜在敞口 (2wk Exposure)	两周潜在敞口占比 (2wk Exposure %)	最大潜在敞口 (Peak Exposure)	最大潜在敞口占比 (Peak Exposure %)
AAA	141	2%	272	2%	462	2%
AA	3,674	59%	7,574	57%	14,990	51%
A	1,347	22%	3,338	25%	10,433	35%
BBB	95	2%	190	1%	502	2%
BBB以下	980	16%	1,972	15%	136	0%
未评级						
Total (总计)	6,237	100%	13,347	100%	29,672	100%

Aggregation at Counterparty Level

- ▶ If there are netting agreements, trades with positive values at time of default offset the ones with negative values with each netting agreement, so that the Cpty-level exposure is given by

$$PE(t) = \sum_k \max \left[0, \sum_{i \in NA_k} V_i(t) \right] + \sum_{i \notin NA} \max [0, V_i(t)]$$



Master Netting Agreement

▶ Time Puts (Mutual Term)

- Give each party the option to get out of transaction after a preset period.

▶ Collateral (CSD Agreement)

- Require each party to post collateral in support of transaction

▶ Margin Threshold

- Create a cash buffer to mitigate settlement risk with a weaker Cpty by altering the cashflows between amount to be received and amount to be paid

▶ Rating Trigger (vs Market-Value Trigger)

- Downgrade triggers allow the bank to pre-settle outstanding transactions if a Cpty's grade fall below a certain level

Netting of Current Exposure

► Objectives

- Quantifying potential future exposure (PFE) at Cpty level
- Estimating capital charge from either default risk or loss of economic value

► Current Exposure

- Under a legally enforceable netting agreement k

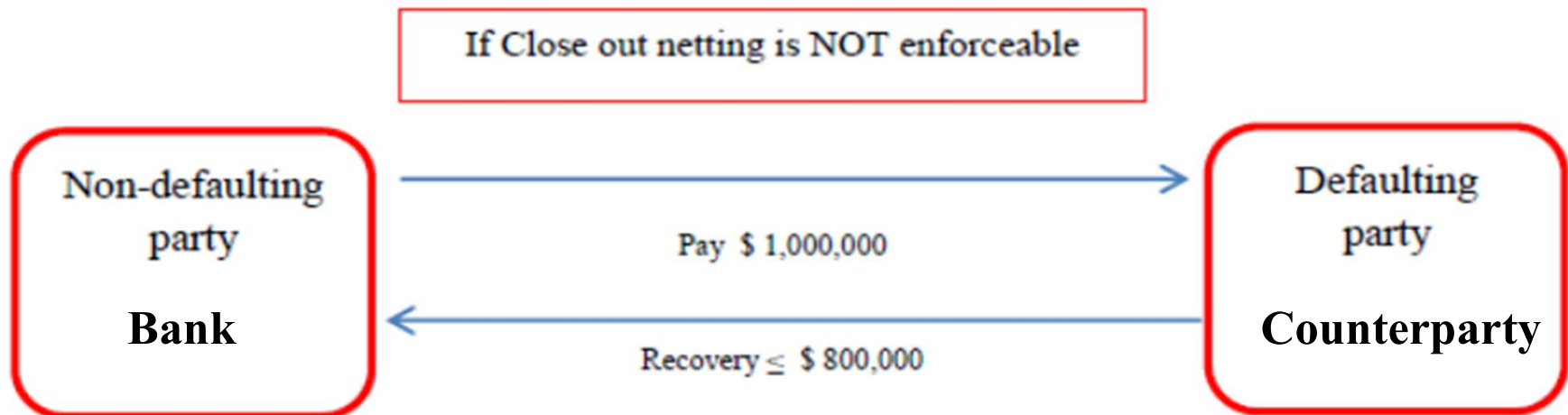
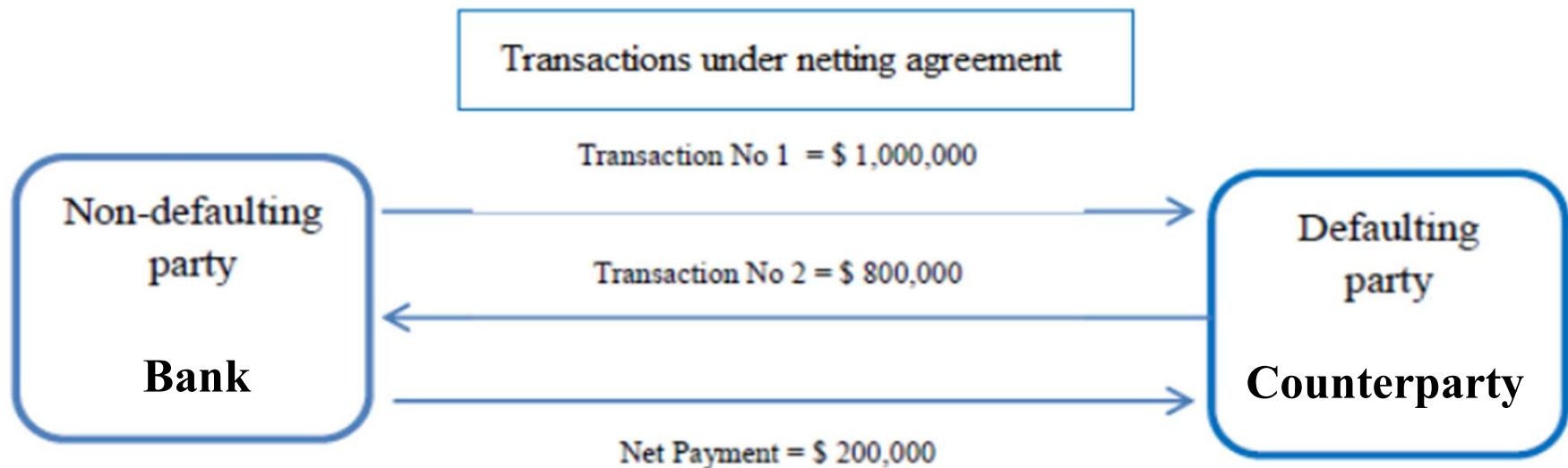
$$CE_{\text{netting } k} = \max \left(0, \sum_j PV_{j,k} \right)$$

- Sum across different netting agreements k and k'

$$CE = CE_{\text{netting } k} + CE_{\text{netting } k'}$$

ISDA Master Agreement

► Close-out Netting



ISDA Master Agreement

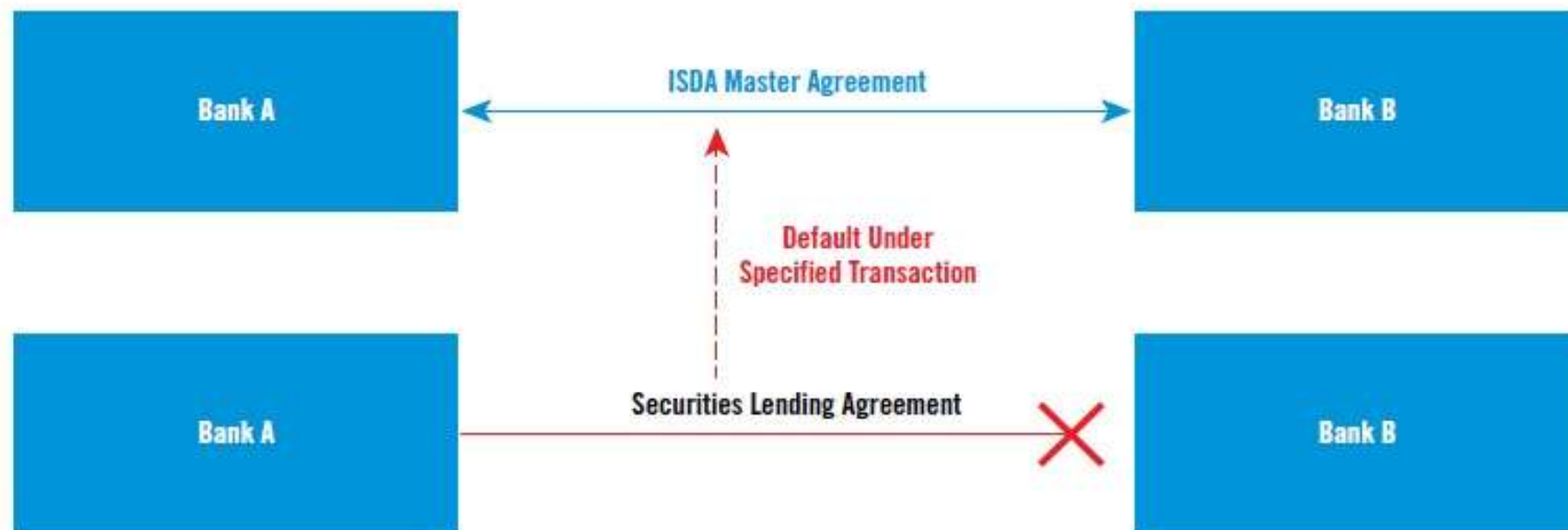
► Default under Specified Transaction

Example of Default under Specified Transaction

Bank A and Bank B enter into a number of agreements, including an ISDA Master Agreement and securities lending agreement (see Figure 3).

While Bank A continues to meet its obligations under the ISDA Master Agreement, it fails to make a payment when due under the securities lending agreement.

Figure 3



Upon this default, an event of default may occur with respect to Bank A under the ISDA Master Agreement.

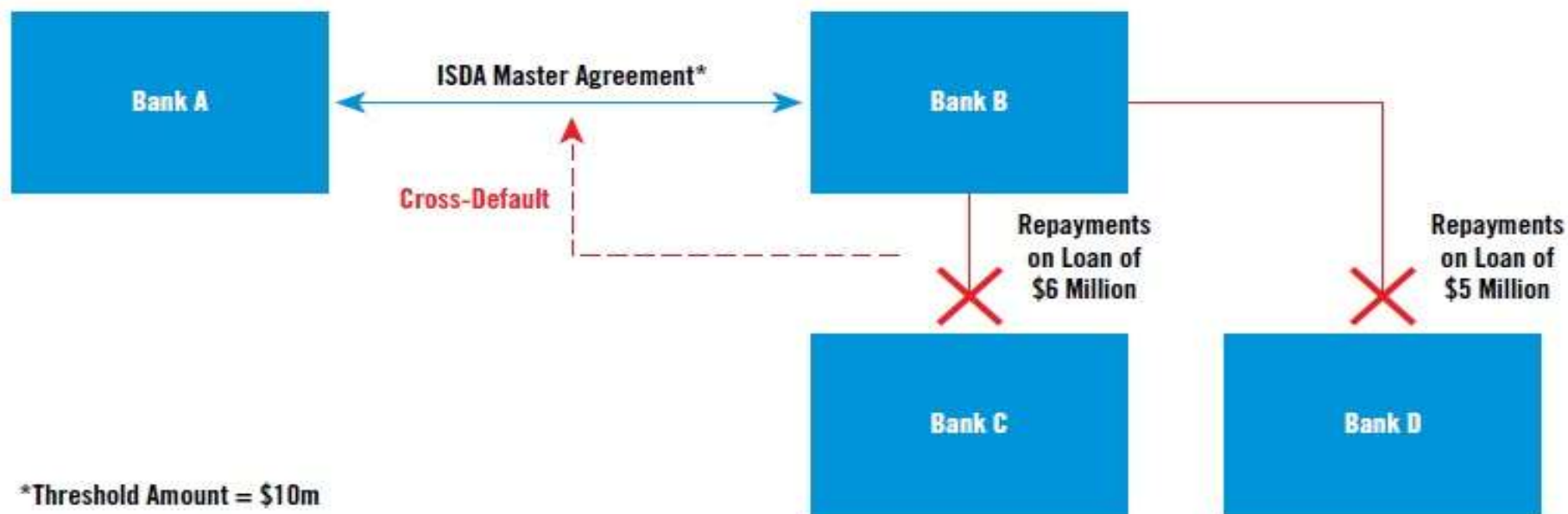
ISDA Master Agreement: Cross Default

Example of Cross-Default

Bank A and Bank B have entered into an ISDA Master Agreement. For the purposes of the cross-default provision, they have agreed that the threshold amount in respect of any specified indebtedness should be \$10 million (see Figure 4).

Bank B also has two loans with Bank C and Bank D totaling \$6 million and \$5 million, respectively. Bank B fails to make repayments under both loans.

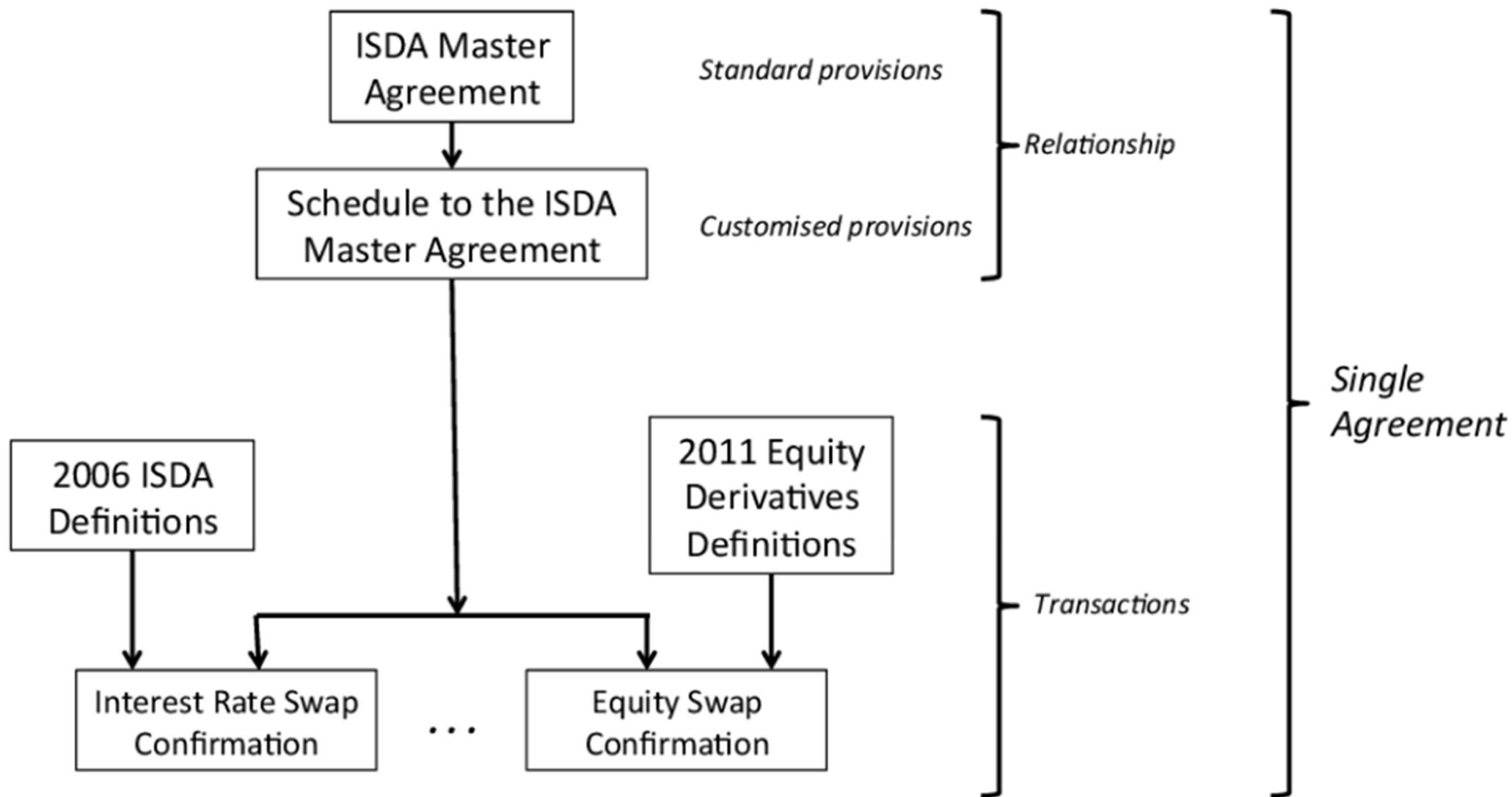
Figure 4



Bank B is therefore in default with respect to an amount of specified indebtedness totaling \$11 million. This is in excess of the \$10 million threshold amount agreed by Bank A and Bank B. This may therefore constitute an event of default in respect of Bank B under the ISDA Master Agreement.

ISDA Document Architecture

► ISDA schedules and product definitions



ISDA Document Template and Sections

Section #	1992 ISDA MA	2002 ISDA MA
1.	Interpretation	Interpretation
2.	Obligations	Obligations
3.	Representations	Representations
4.	Agreements	Agreements
5.	Events of Default and Termination Events	Events of Default and Termination Events
6.	Early Termination	Early Termination; Close-Out Netting
7.	Transfer	Transfer
8.	Contractual Currency	Contractual Currency
9.	Miscellaneous	Miscellaneous
10.	Offices; Multibranch Parties	Offices; Multibranch Parties
11.	Expenses	Expenses
12.	Notices	Notices
13.	Governing Law and Jurisdiction	Governing Law and Jurisdiction
14.	Definitions	Definitions